

Chapter4: Counter and DAC & ADC Part-I

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Objective

- Recognize type of flipflops and realize difference between rising edge/falling edge trigger.
- Clarify mechanism inside a counter circuit and express technique of counting up/down.
- Explain and illustrate concept of convert analog signal to digital signal.

Topic

- Combinational logic vs Sequential logic circuit
- Rising/Falling edge trigger
- JK-flipflop
- D-flipflop
- T-flipflop
- Frequency division circuit
- First-in first-out circuit
- Counter circuit
- Analog signal to Digital signal concept

World of digital logic circuit

Combinational
logic circuit

Sequential
logic circuit

Key characteristics

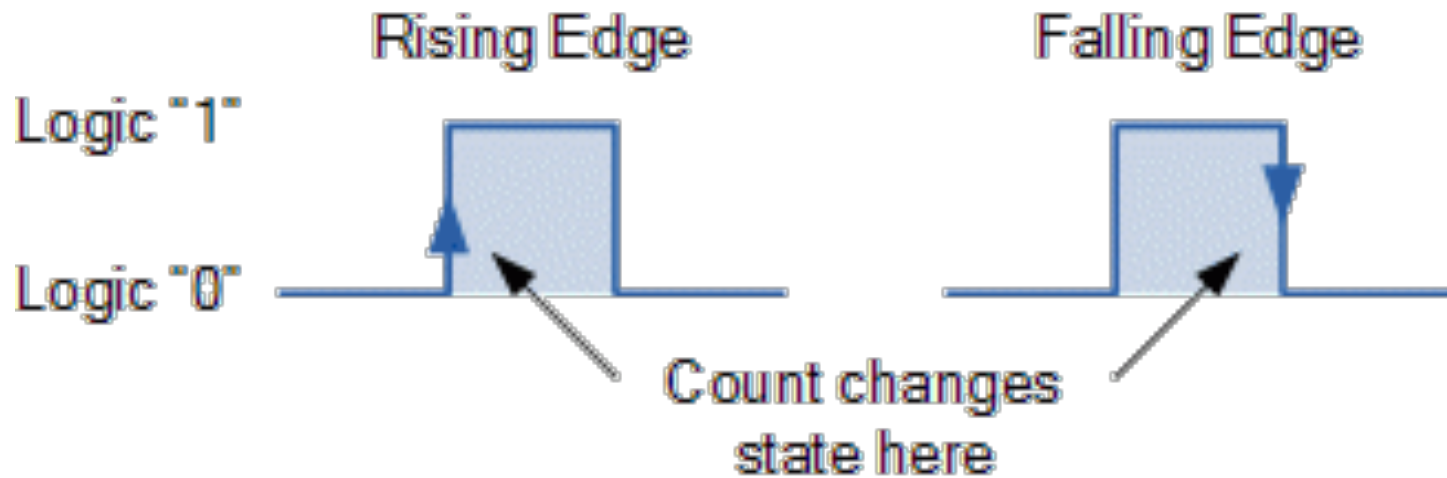
Combinational logic circuit

- Memoryless
- Output instant in time
- No clock signal

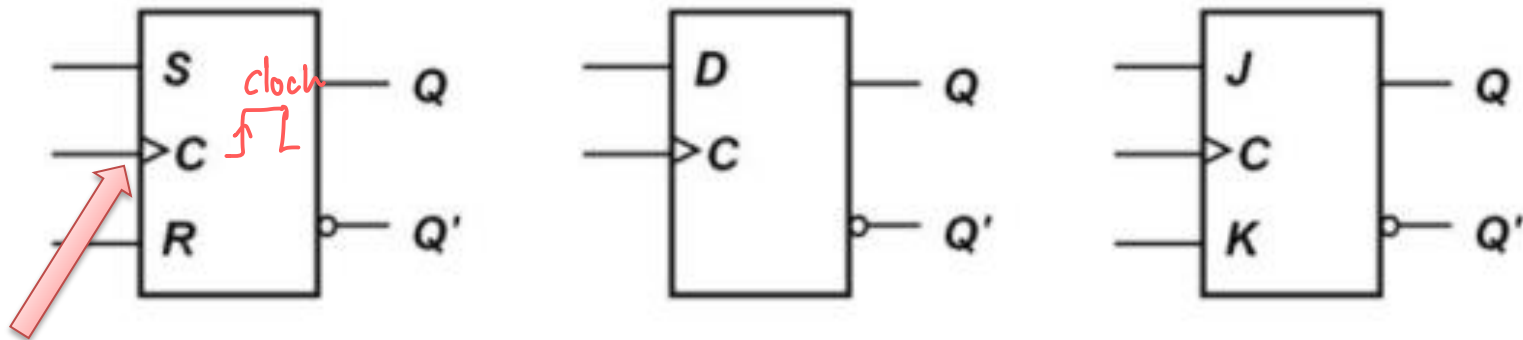
Sequential logic circuit

- Memory
- Output depends on previous states
- Related clock signal

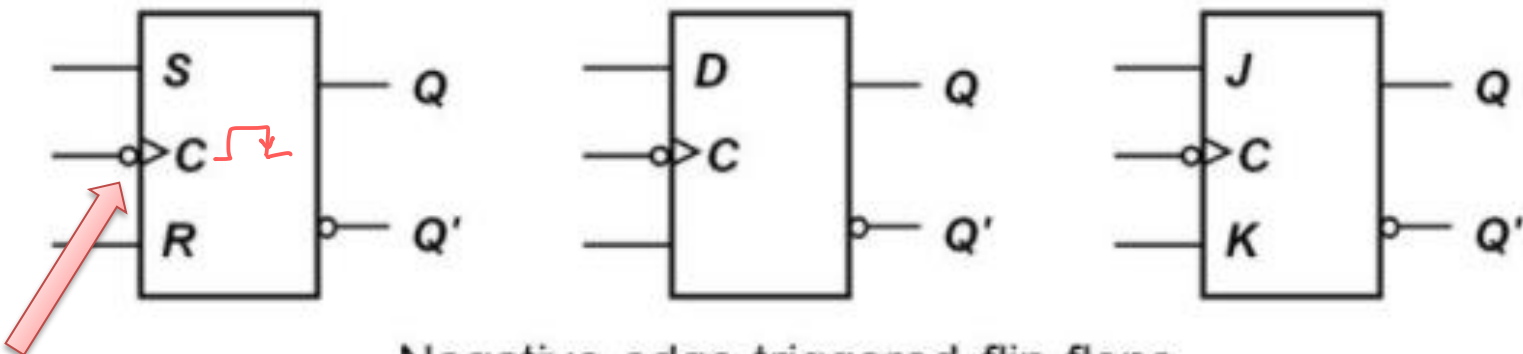
Rising/Falling edge trigger



Flip-flop symbols in rising/falling edge trigger



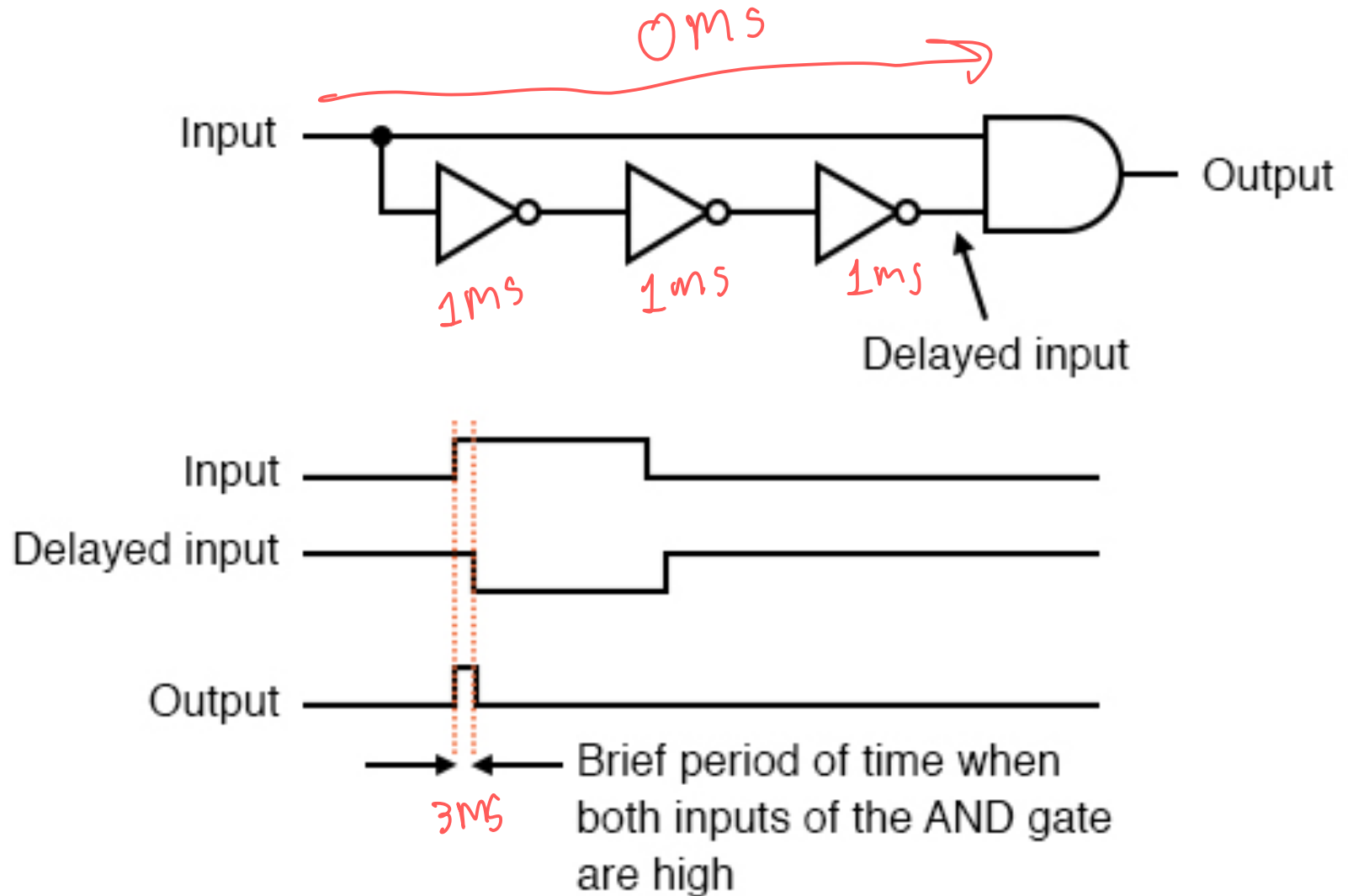
Positive edge-triggered flip-flops



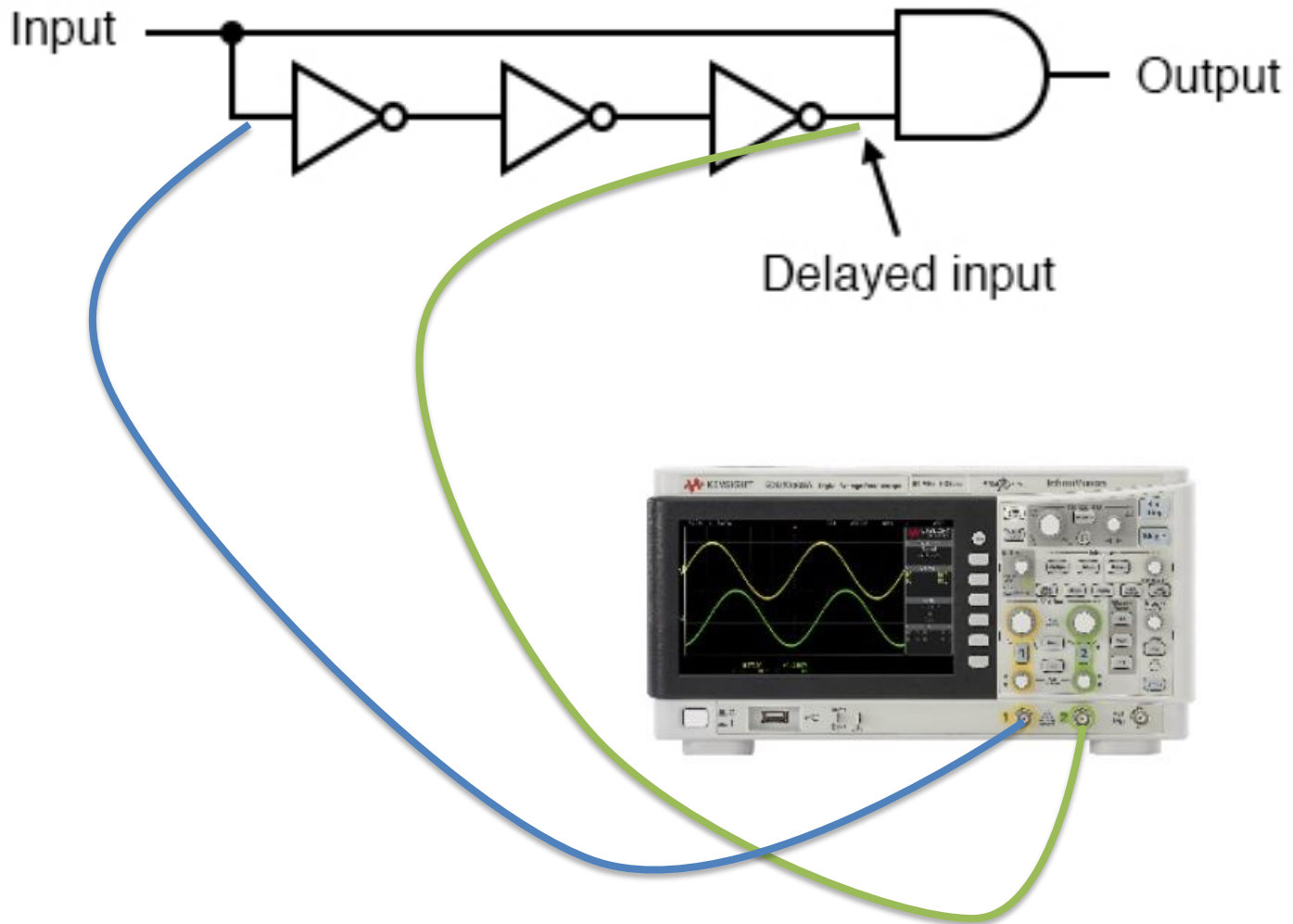
Negative edge-triggered flip-flops

Some textbook for rising/falling edge trigger called positive/negative edge trigger

Circuit detects positive edge trigger

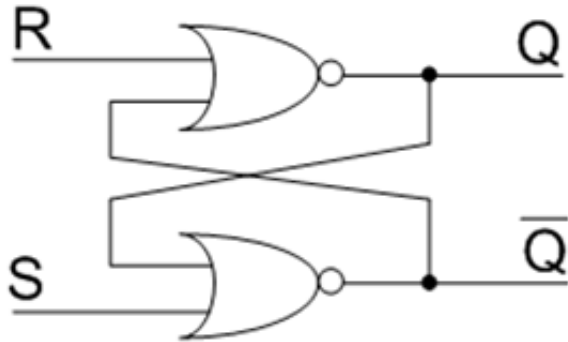


Activity 4.1 Delay time measurement in the positive-edge trigger circuit

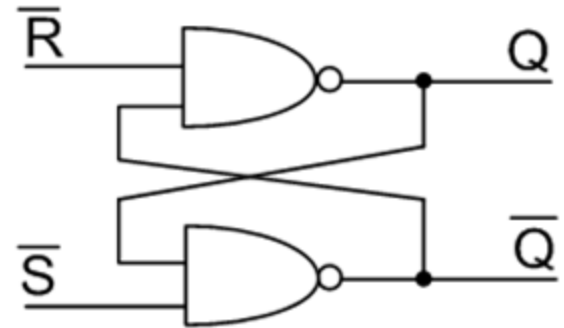


Set-Reset Flip-flop (SR-flipflop)

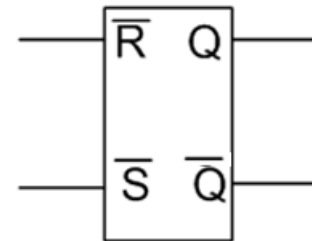
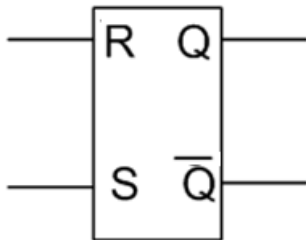
SR flipflop is two types by the structure of logic gate.



NOR gate SR-flipflop

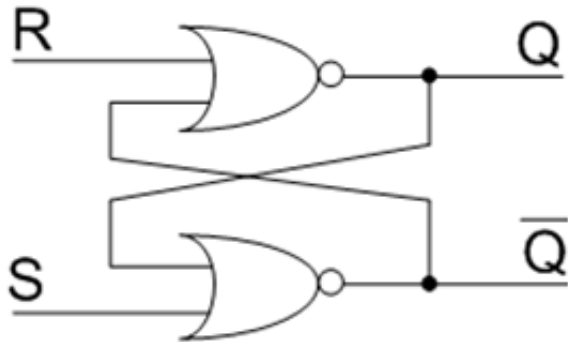


NAND gate SR-flipflop



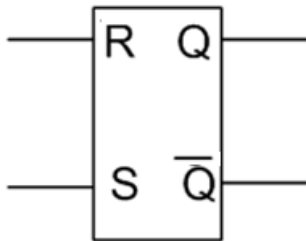
Set-Reset Flip-flop (SR-flipflop)

Truth table



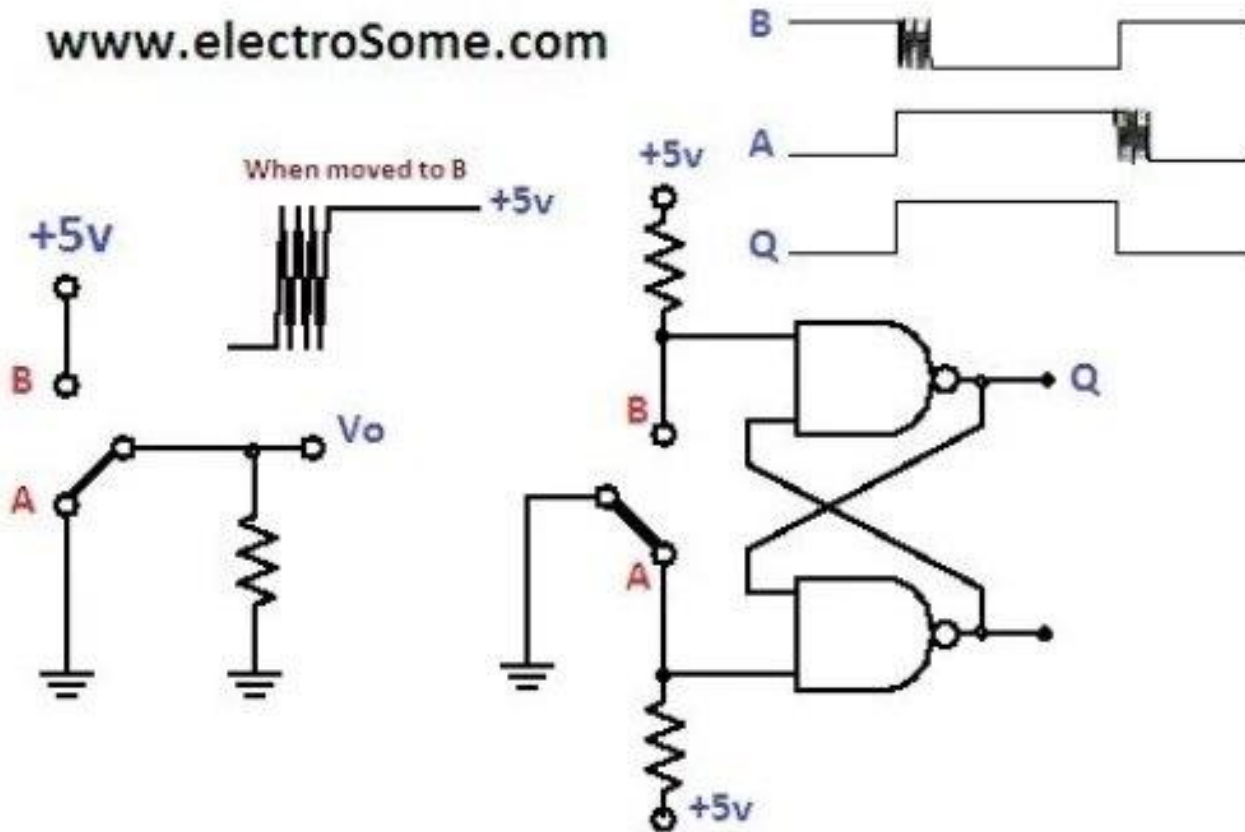
S	R	Q
0	0	No change
0	1	1 = Set
1	0	0 = Reset
1	1	Restrict Combination

NOR gate SR-flipflop

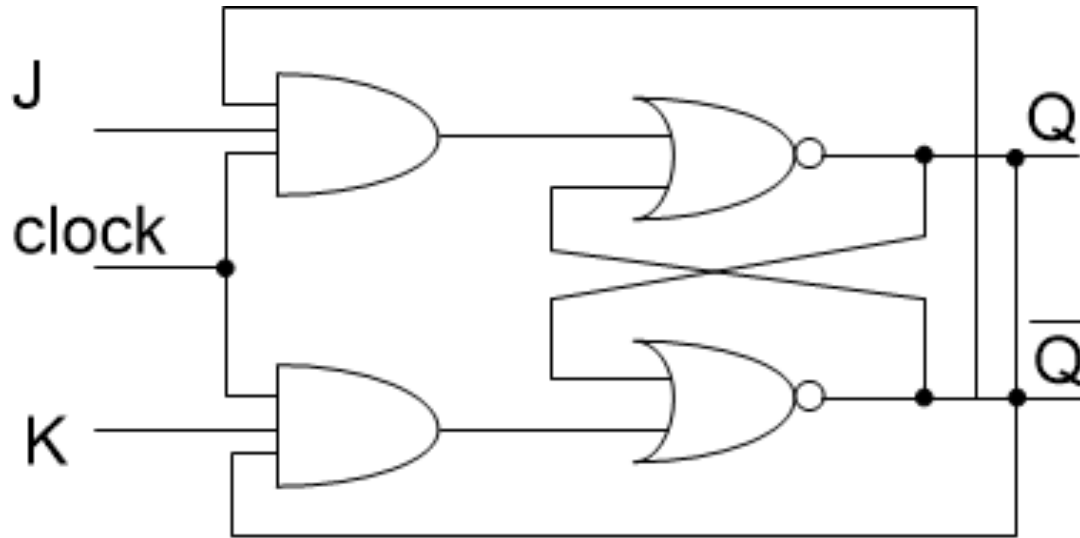


Example application uses SR-flipflop

- Debounced switch

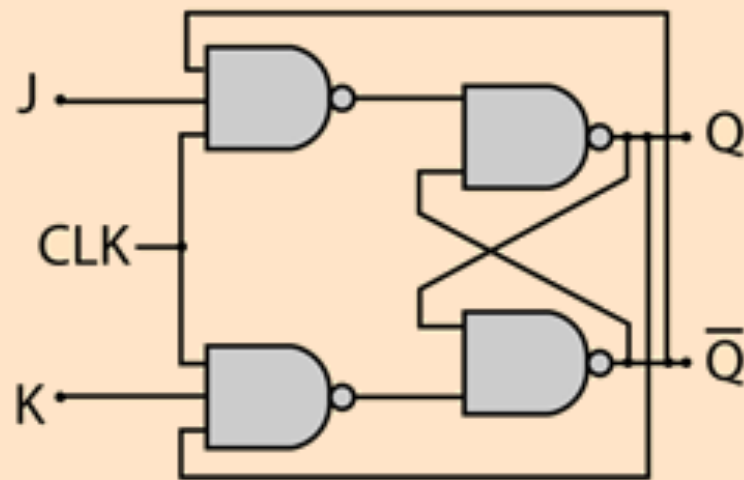


JK Flip-flop

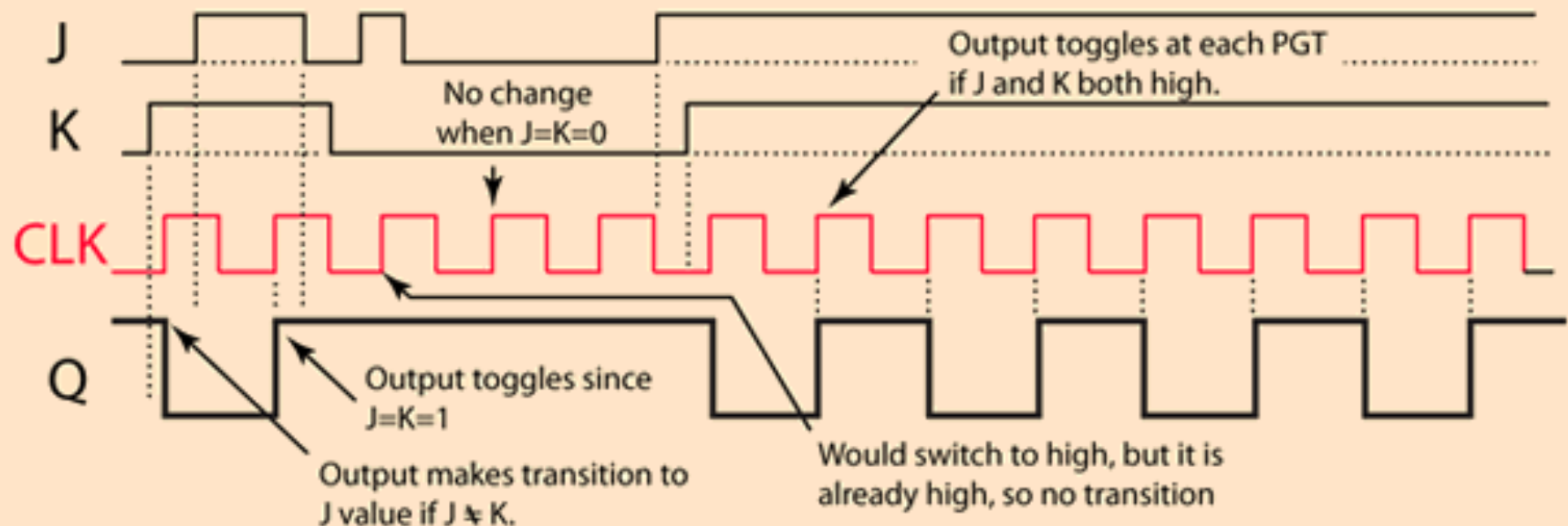


- Invented by Jack Kilby
- The JK flip-flop modified from SR flip-flop by adding a clock input to prevent the invalid output condition that occurs when both S and R are equal to logic "1".

Switching Example: J-K Flip-Flop



The positive going transition (PGT) of the clock enables the switching of the output Q. The "enable" condition does not persist through the entire positive phase of the clock. The J & K inputs alone cannot cause a transition, but their values at the time of the PGT determine the output according to the [truth table](#). This is an application of the versatile [J-K flip-flop](#).



JK Flip-flop Truth table

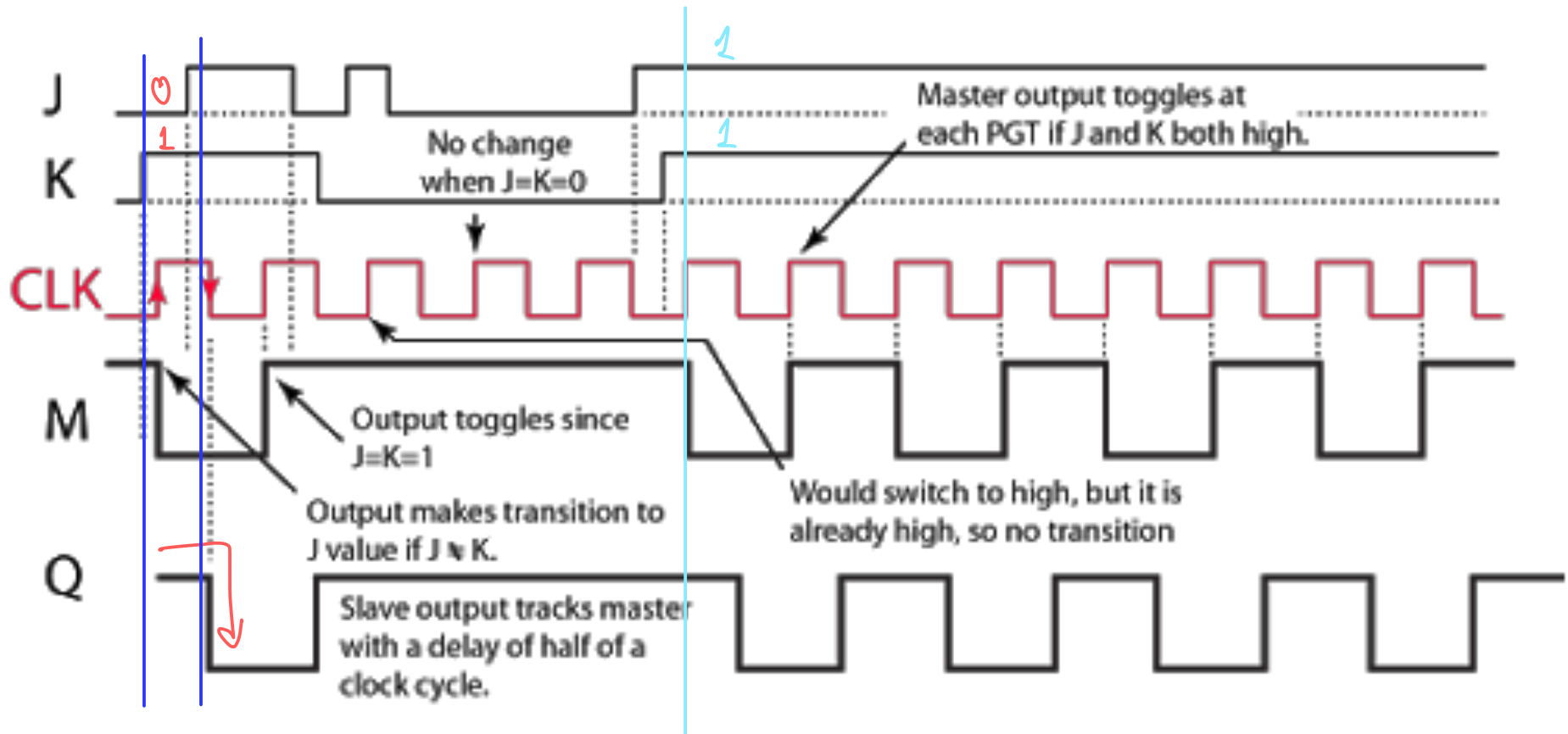
Clk	J	K	Q	Description
No clock	?	?	?	No change
	0	0	?	No change
	0	1	?->0	Reset
	1	0	?->1	Set
	1	1	0->1 1->0	Toggle

Handwritten red text: } 0 1 0 1

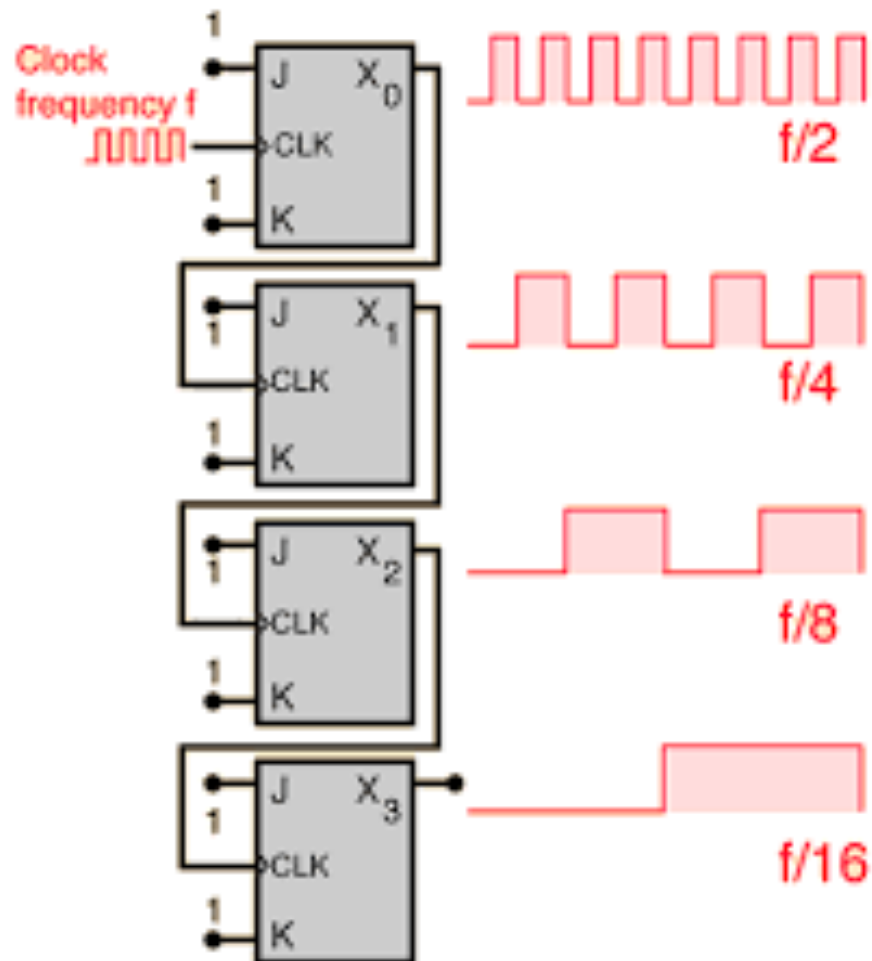
Handwritten red text: 0 → 0
1 → 0

Handwritten red text: 1 → 1
0 → 1

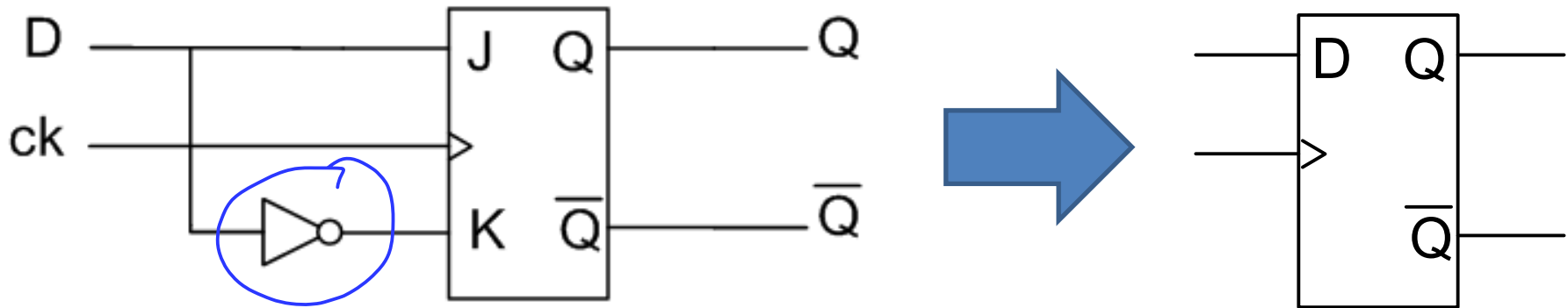
JK flip-flop in timing states



Application used JK flip-flop



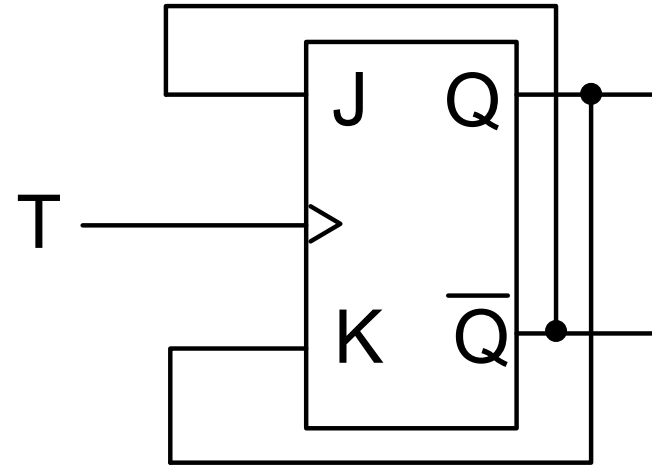
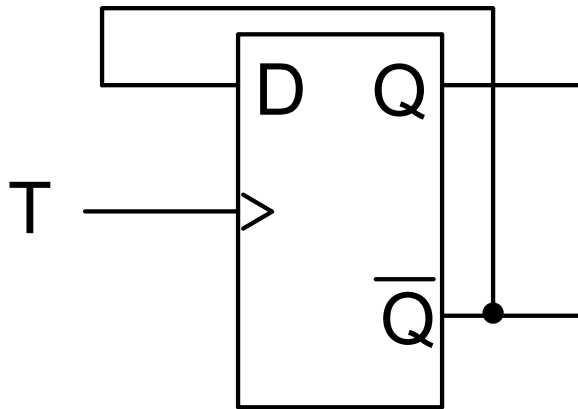
Making D flip-flop from JK



Clk	J	K	Q	Description
No clock	?	?	?	No change
	0	1	?->0	Store "0"
	1	0	?->0/1	Store "1"

T Flip-flop from D and JK

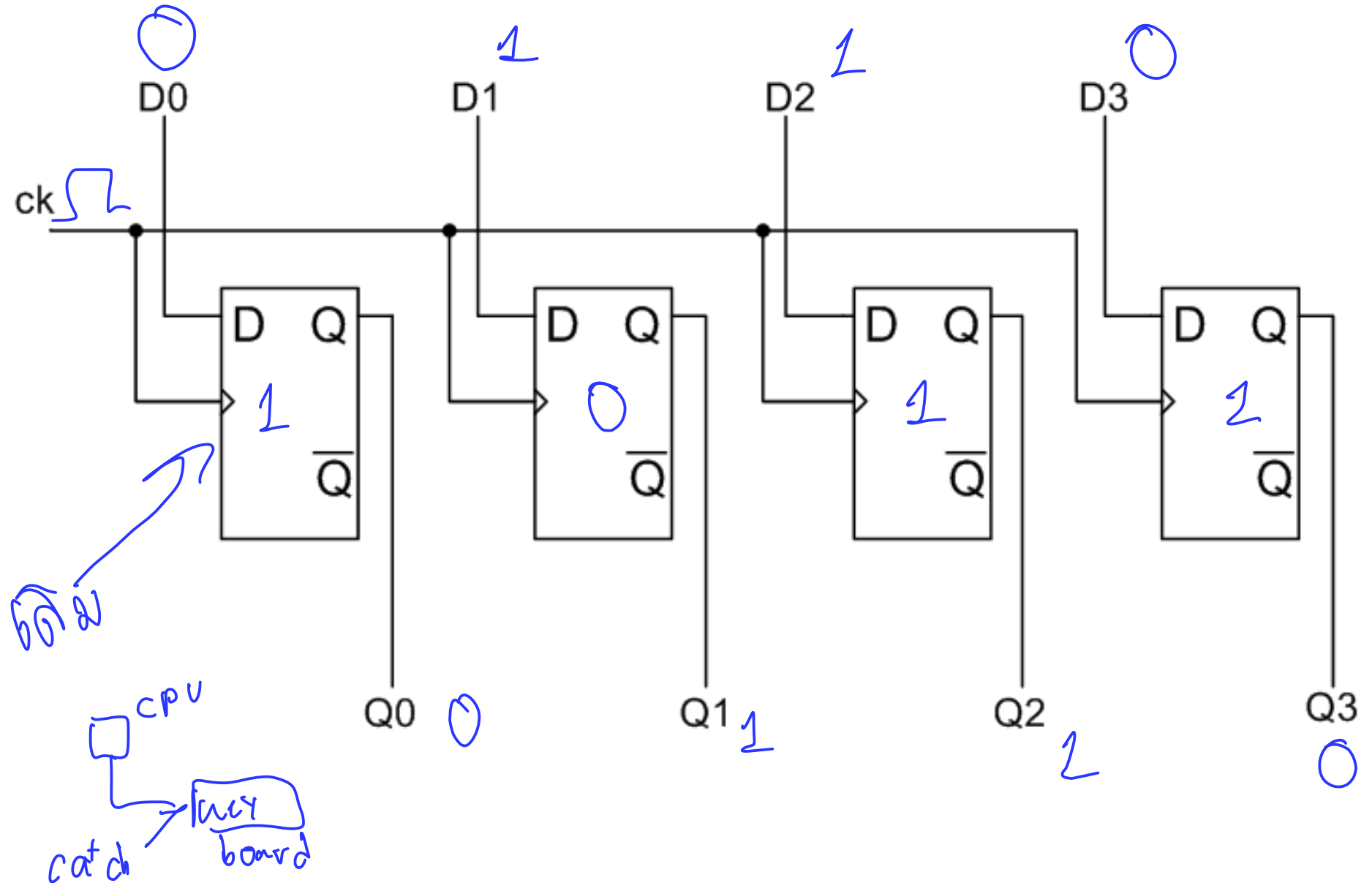
↓
Data



Clk (T)	Q 	Description
No clock	?	No change
	?->0	store "0"

toggle

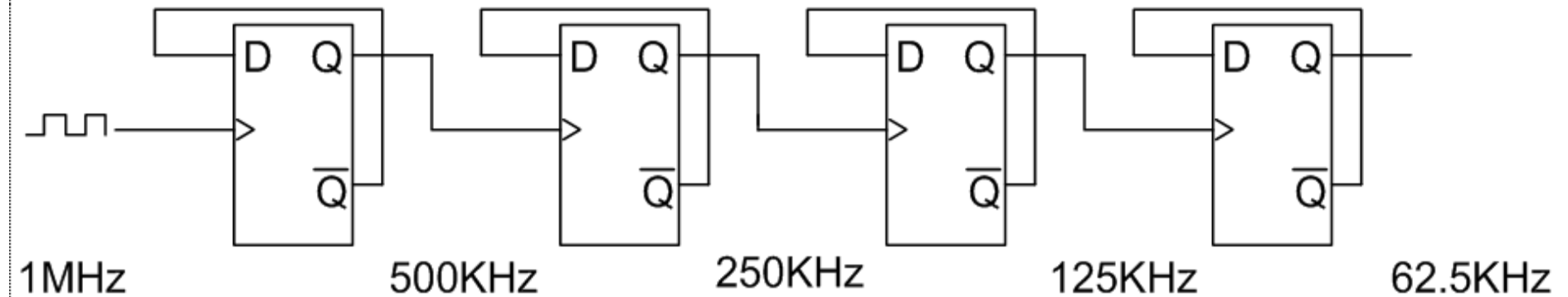
4-bit Latch circuit



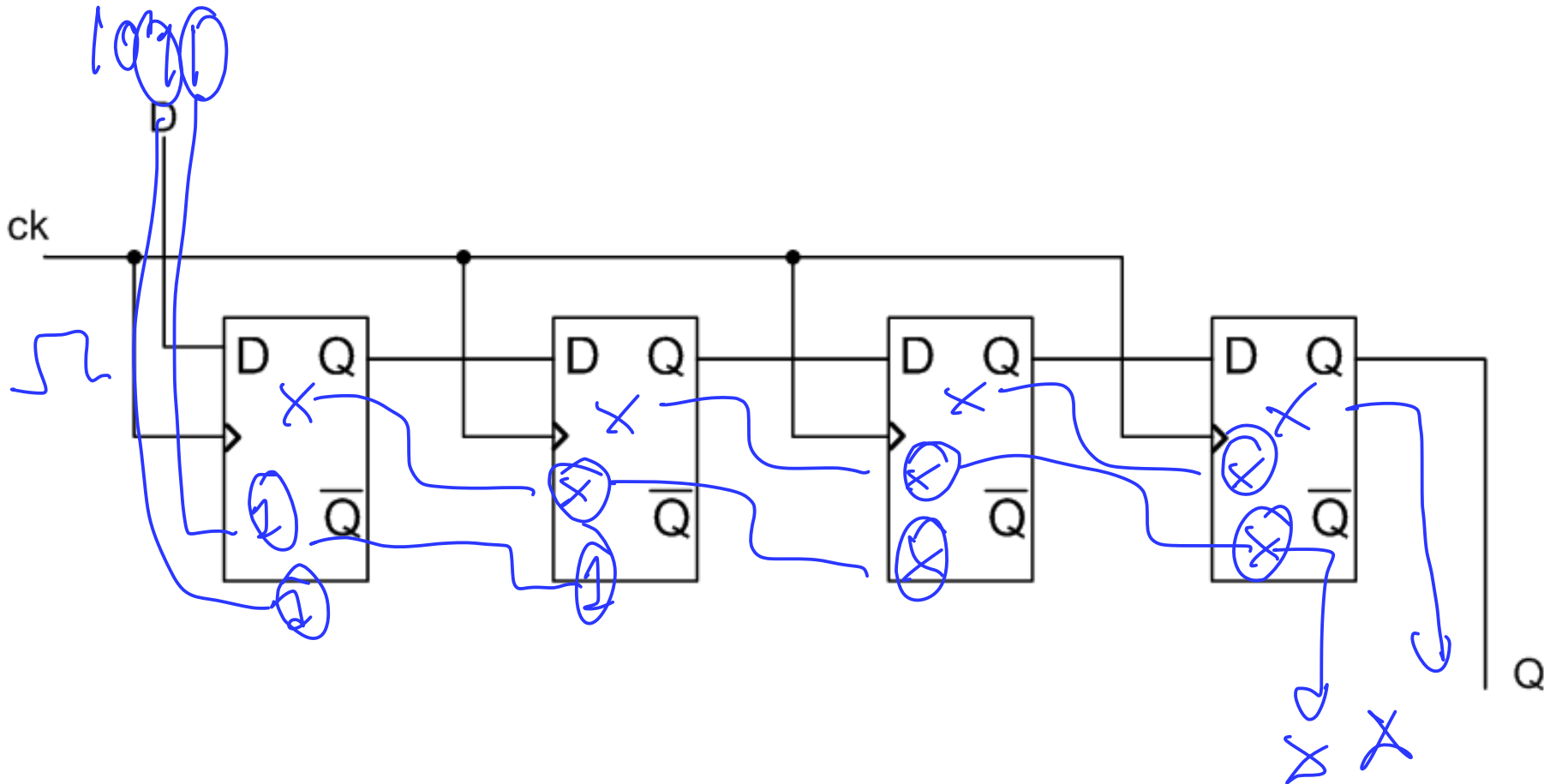
Frequency division circuit

— [0] —
↖

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4-bit First-in First-out (FIFO)



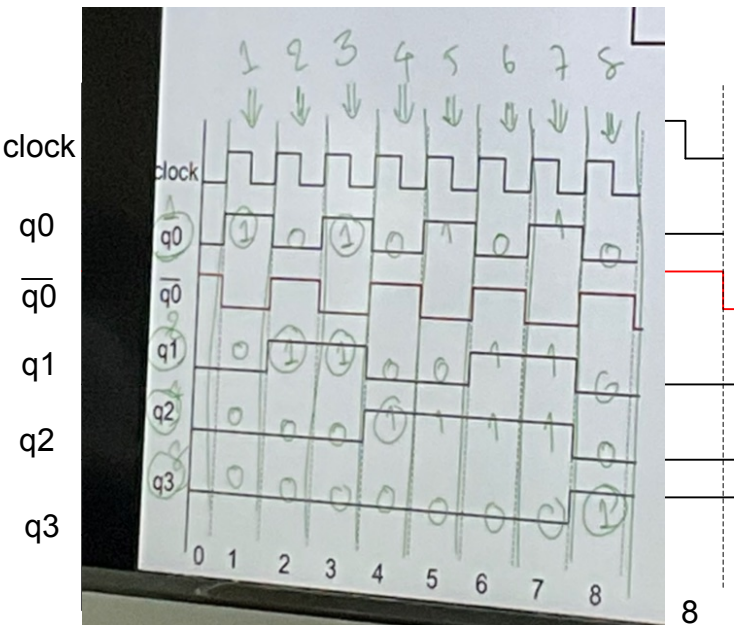
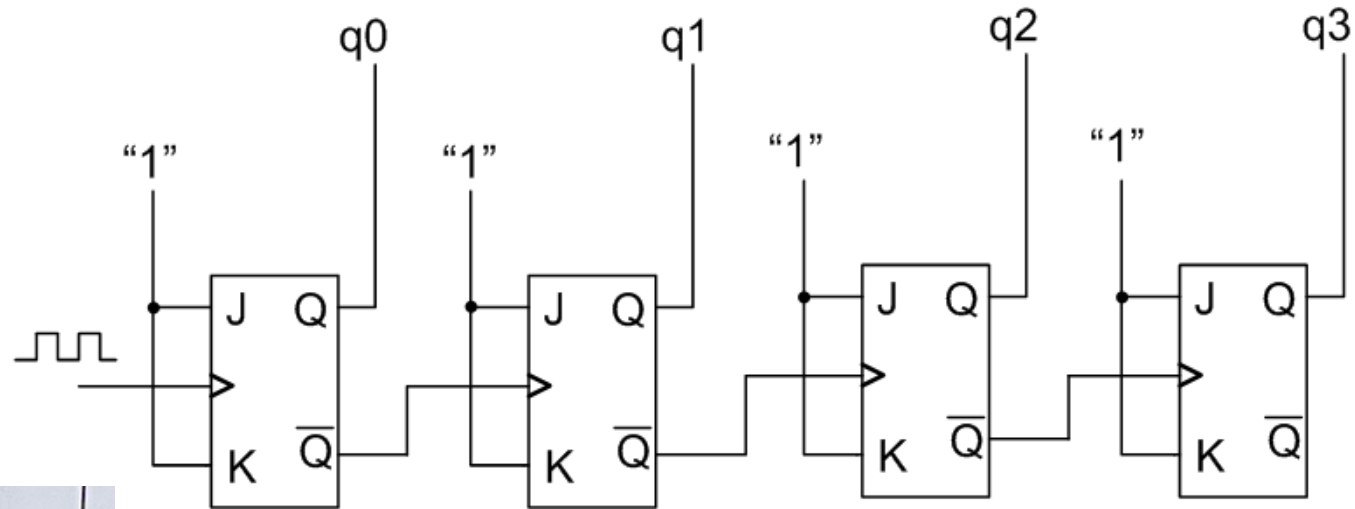
Counter circuit

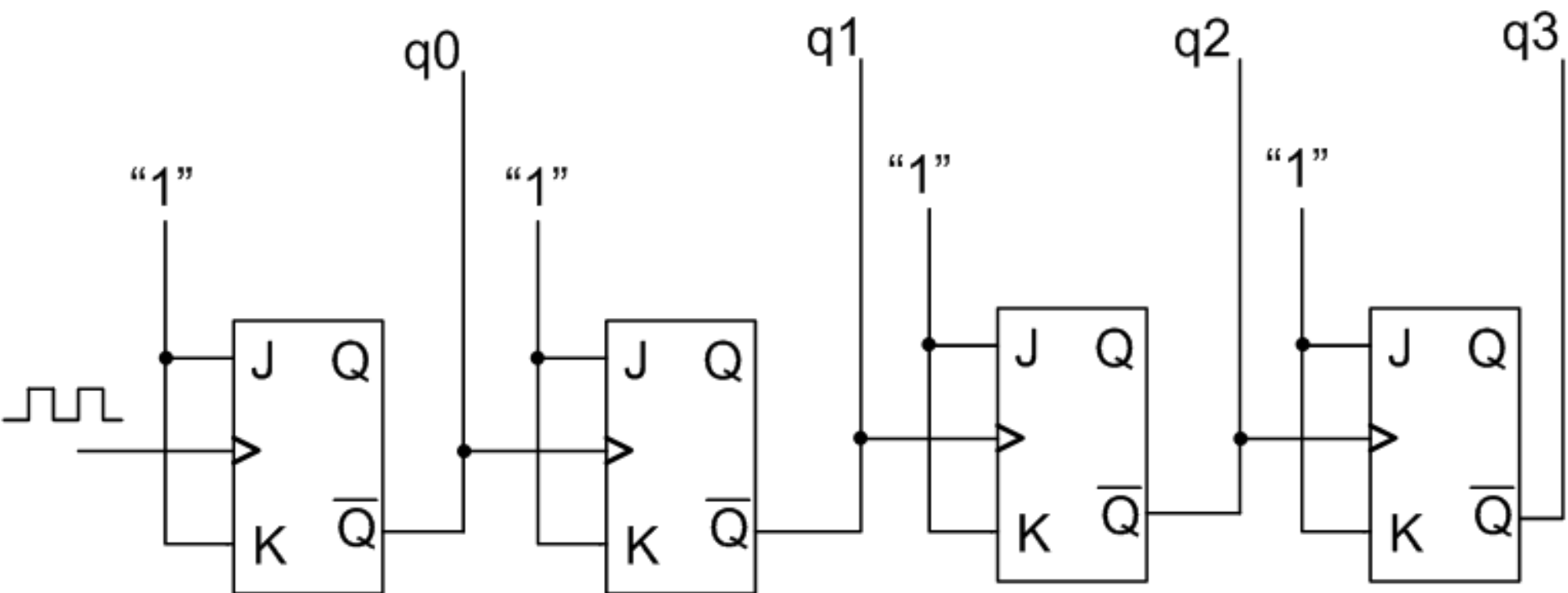
- The digital circuit has divided type of counter circuit by focusing at a characteristic of input clock to the gate in the circuit.

Asynchronous
circuit

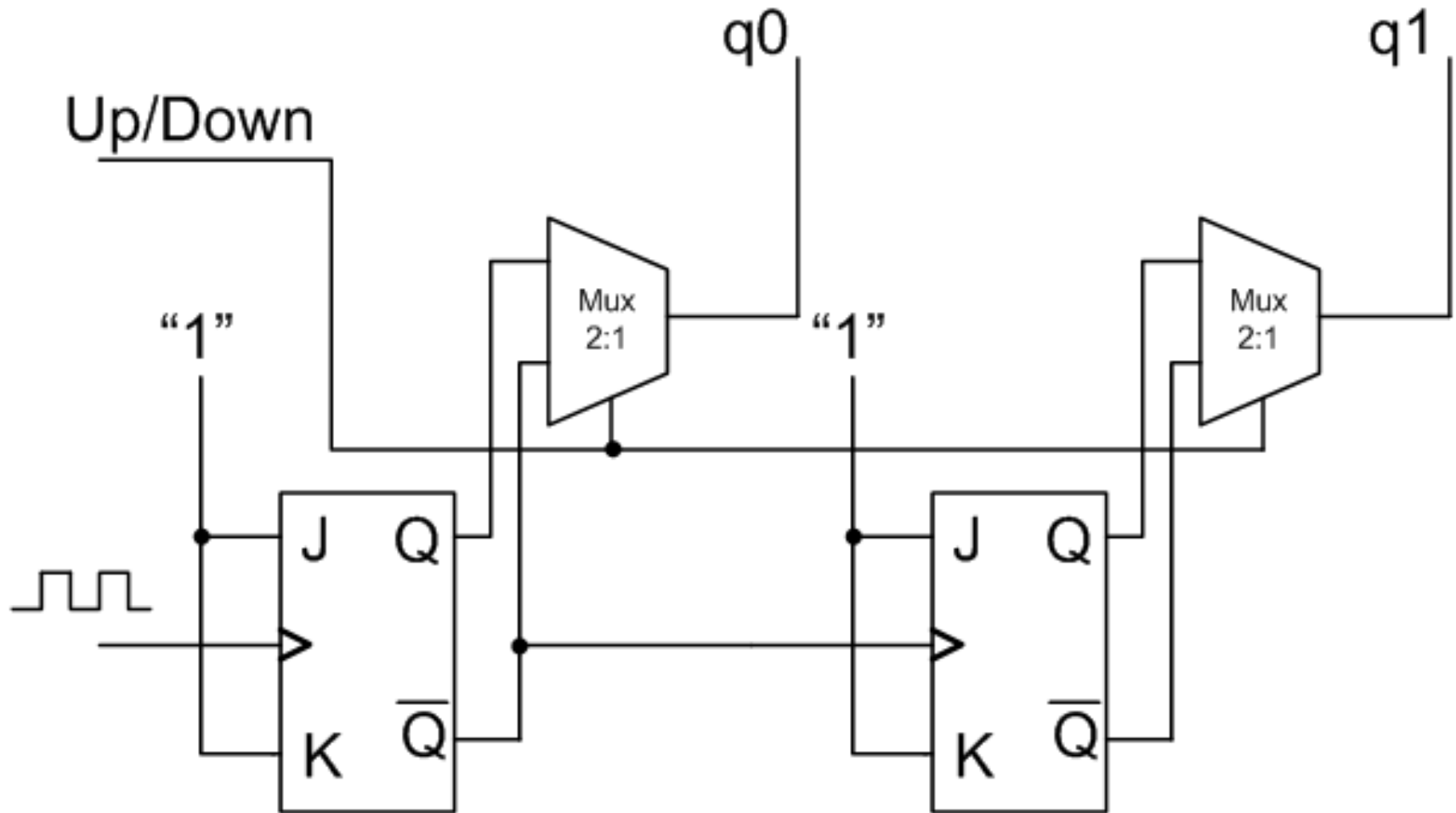
Synchronous
circuit

Asynchronous counter





2-bit Asynchronous count up/down circuit



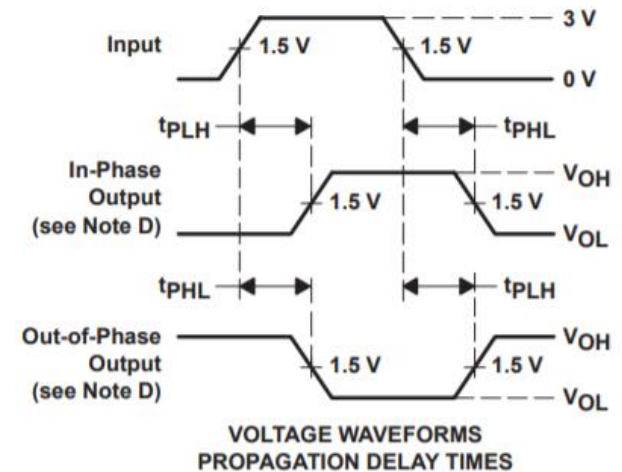
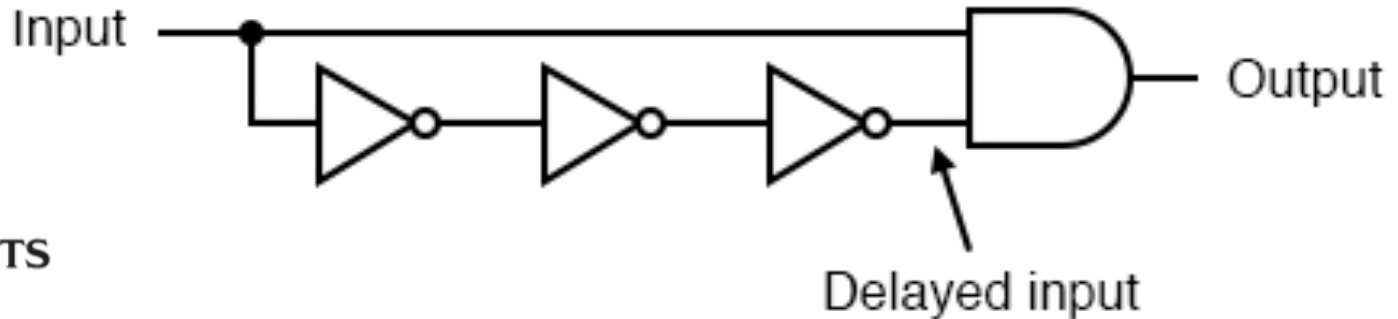
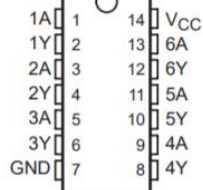
Delay-time problem in asynchronous counter circuit



**SN5404, SN54LS04, SN54S04,
SN7404, SN74LS04, SN74S04
HEX INVERTERS**

SDLS029C - DECEMBER 1983 - REVISED JANUARY 2004

SN5404 . . . J PACKAGE
SN54LS04, SN54S04 . . . J OR W PACKAGE
SN7404, SN74LS04 . . . D, N, OR NS PACKAGE
SN74LS04 . . . D, DB, N, OR NS PACKAGE
(TOP VIEW)



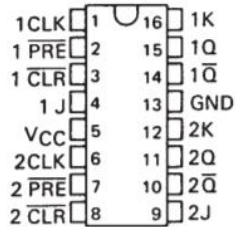
switching characteristics, $V_{CC} = 5\text{ V}$, $T_A = 25^\circ\text{C}$ (see Figure 1)

PARAMETER	FROM (INPUT)	TO (OUTPUT)	TEST CONDITIONS	SN54S04 SN74S04			UNIT
				MIN	TYP	MAX	
t_{PLH}	A	Y	$R_L = 280\ \Omega$, $C_L = 15\text{ pF}$		3	4.5	ns
t_{PHL}					3	5	

Asynchronous counter issue



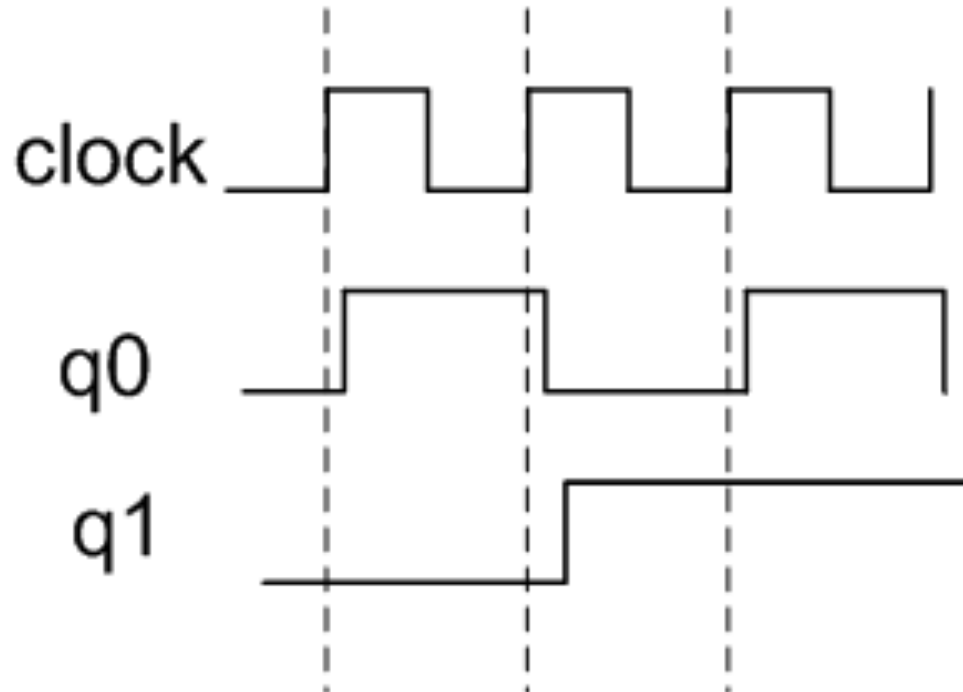
(TOP VIEW)



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FUNCTION TABLE

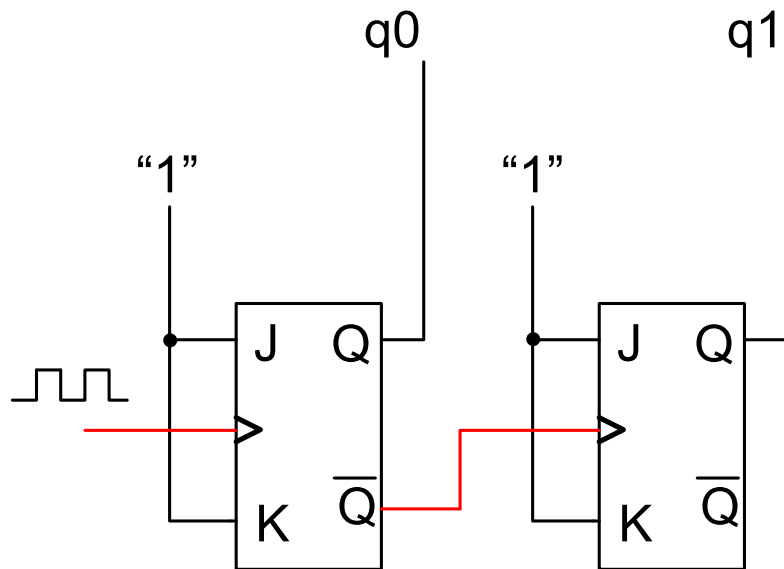
INPUTS					OUTPUTS	
PRE	CLR	CLK	J	K	Q	Q̄
L	H	X	X	X	H	L
H	L	X	X	X	L	H
L	L	X	X	X	H [†]	H [†]
H	H		L	L	Q ₀	Q̄ ₀
H	H		H	L	H	L
H	H		L	H	L	H
H	H		H	H	TOGGLE	



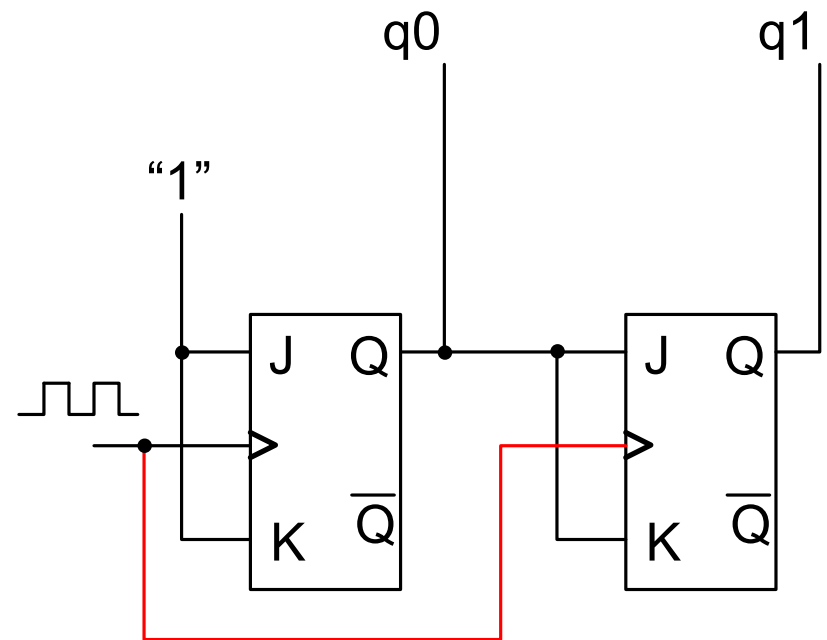
switching characteristics, $V_{CC} = 5\text{ V}$, $T_A = 25^\circ\text{C}$ (see note 3)

PARAMETER	FROM (INPUT)	TO (OUTPUT)	TEST CONDITIONS	MIN	TYP	MAX	UNIT
f _{max}			R _L = 2 kΩ, C _L = 15 pF	30	45		MHz
t _{PLH}	PRE, CLR or CLK	Q or Q̄			15	20	ns
t _{PHL}					15	20	ns

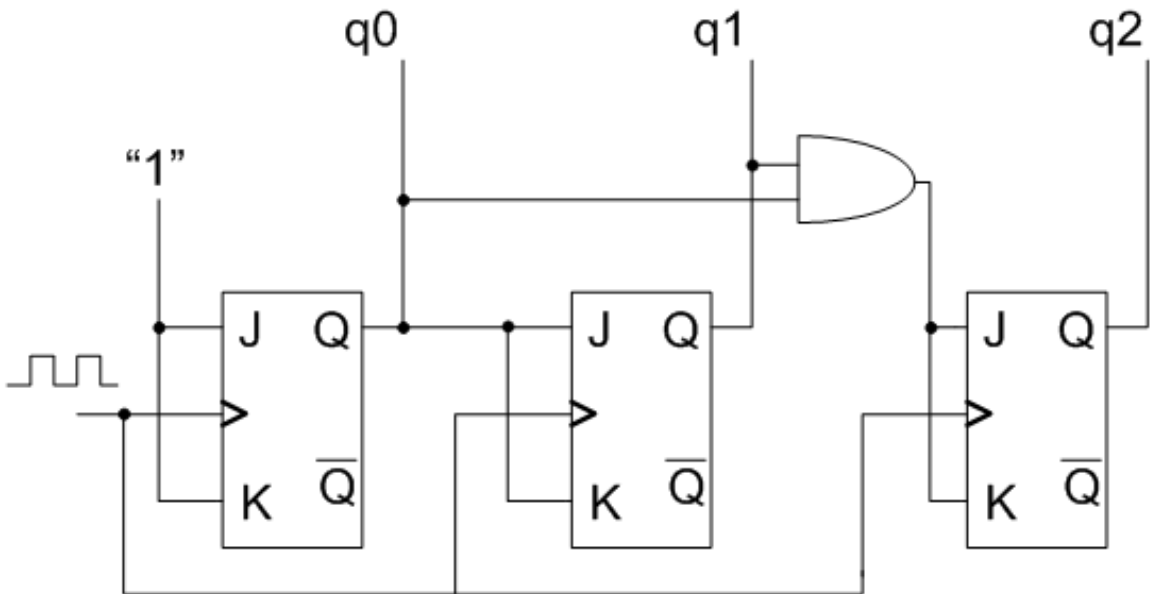
Asynchronous vs Synchronous counter



- Clock or control signal getting from previous states
- Easy design the counter circuit.
- Not suitable for a critical time system.



- A clock signal controls all states
- Complexity for design counter circuit.
- Proper for the critical time system



ลำดับ ck	Q2	Q1	Q0	อธิบาย
1	0	0	0	นับ 0
2	0	0	1	นับ 1
3	0	1	0	นับ 2
4	0	1	1	นับ 3 เอา q0 and q1 = 1
5	1	0	0	นับ 4 เกิด toggle ที่ q2
6	1	0	1	นับ 5
7	1	1	0	นับ 6
8	1	1	1	นับ 7 เอา q0 and q1 = 1
9	0	0	0	นับ 0 เกิด toggle ที่ q2

Analog signal vs Digital signal

Analog Signal



Digital Signal

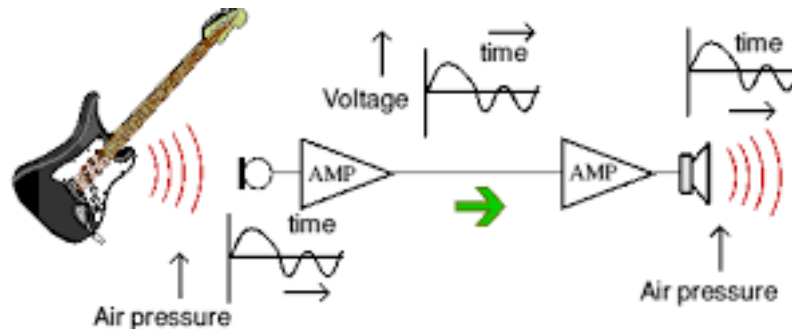


Analog signal vs Digital signal

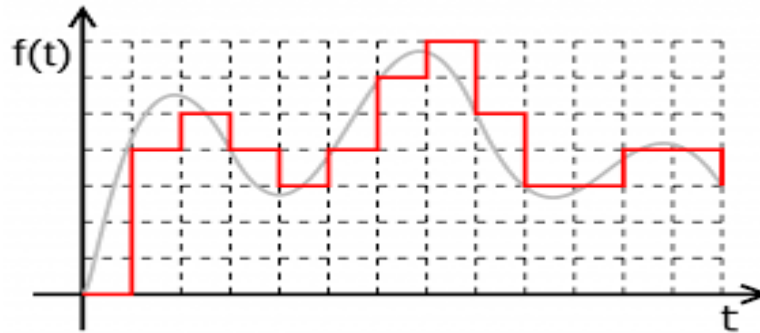
Analog Signal



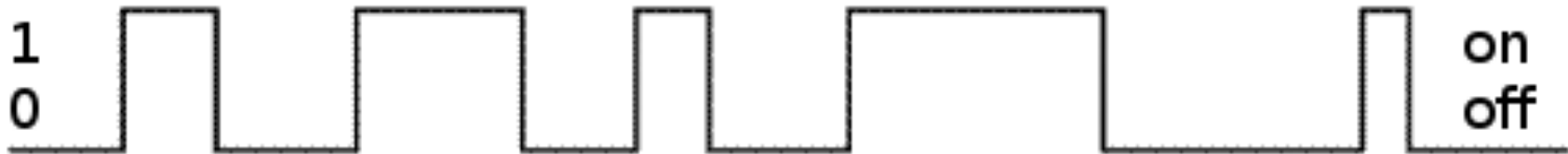
Analog signals are continuous-time signals changing the wave from continuously.



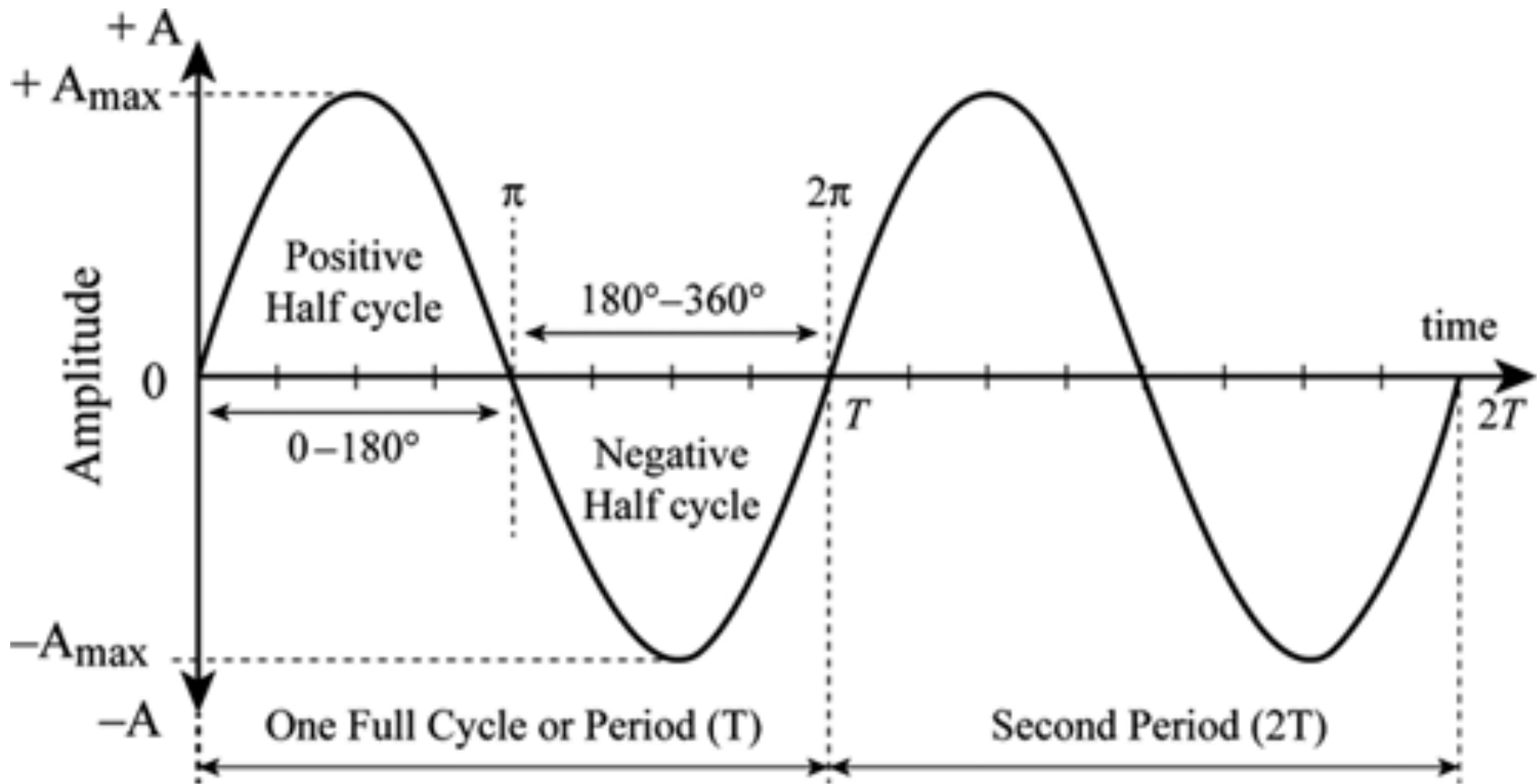
Digital signals are discrete-time signals limited by number of **bits** and the **sampling rate**.



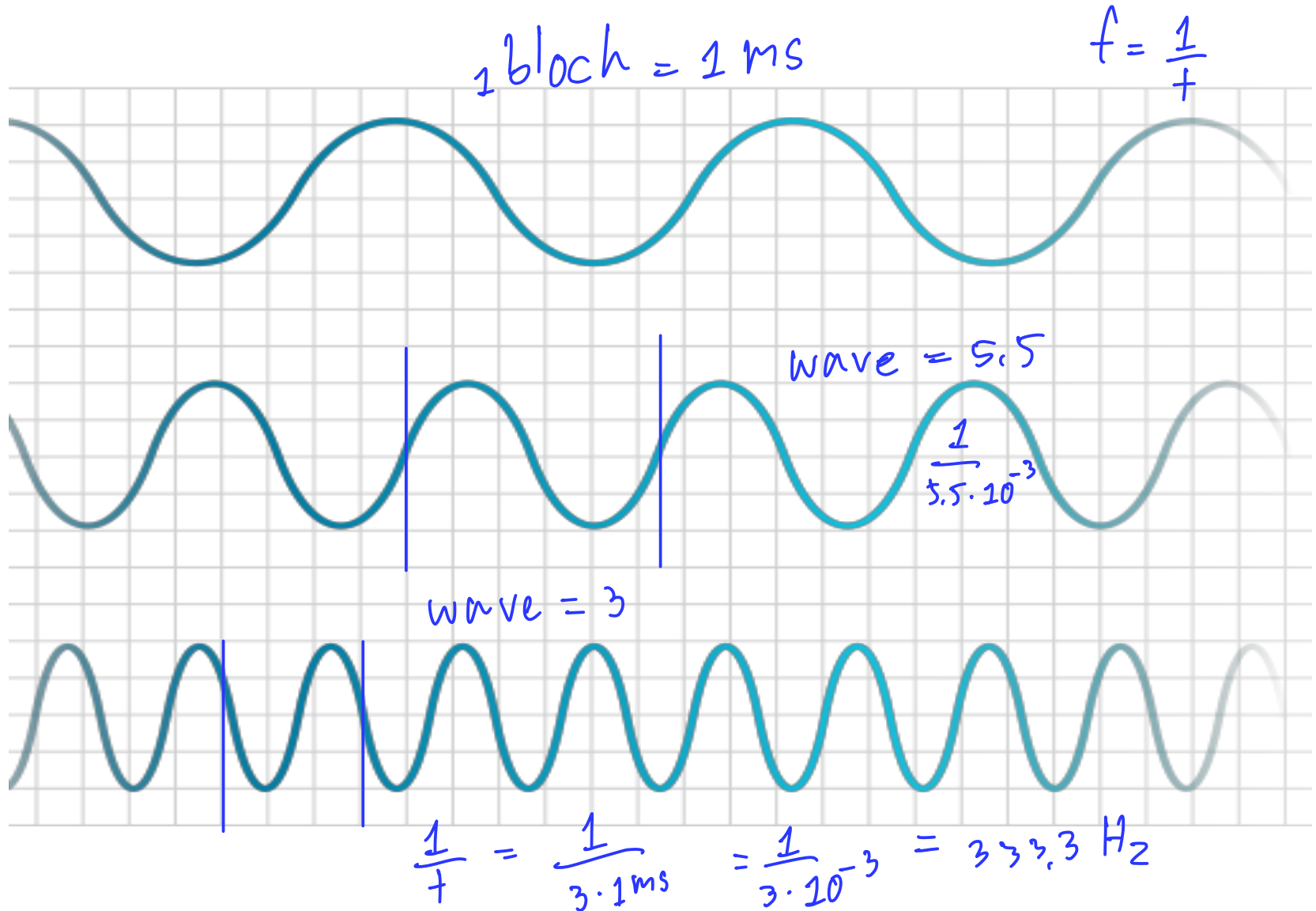
Digital Signal



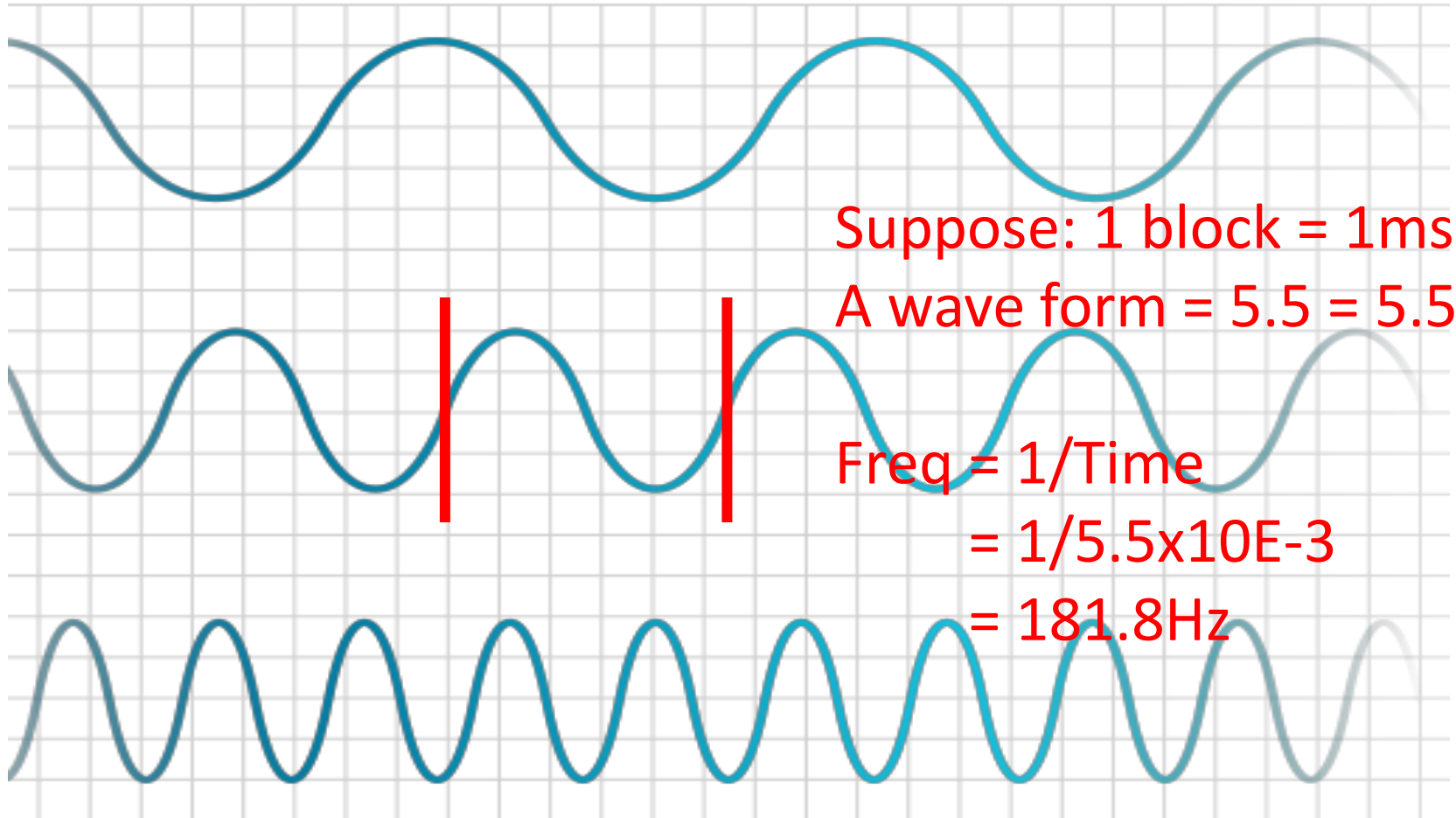
Signal Property



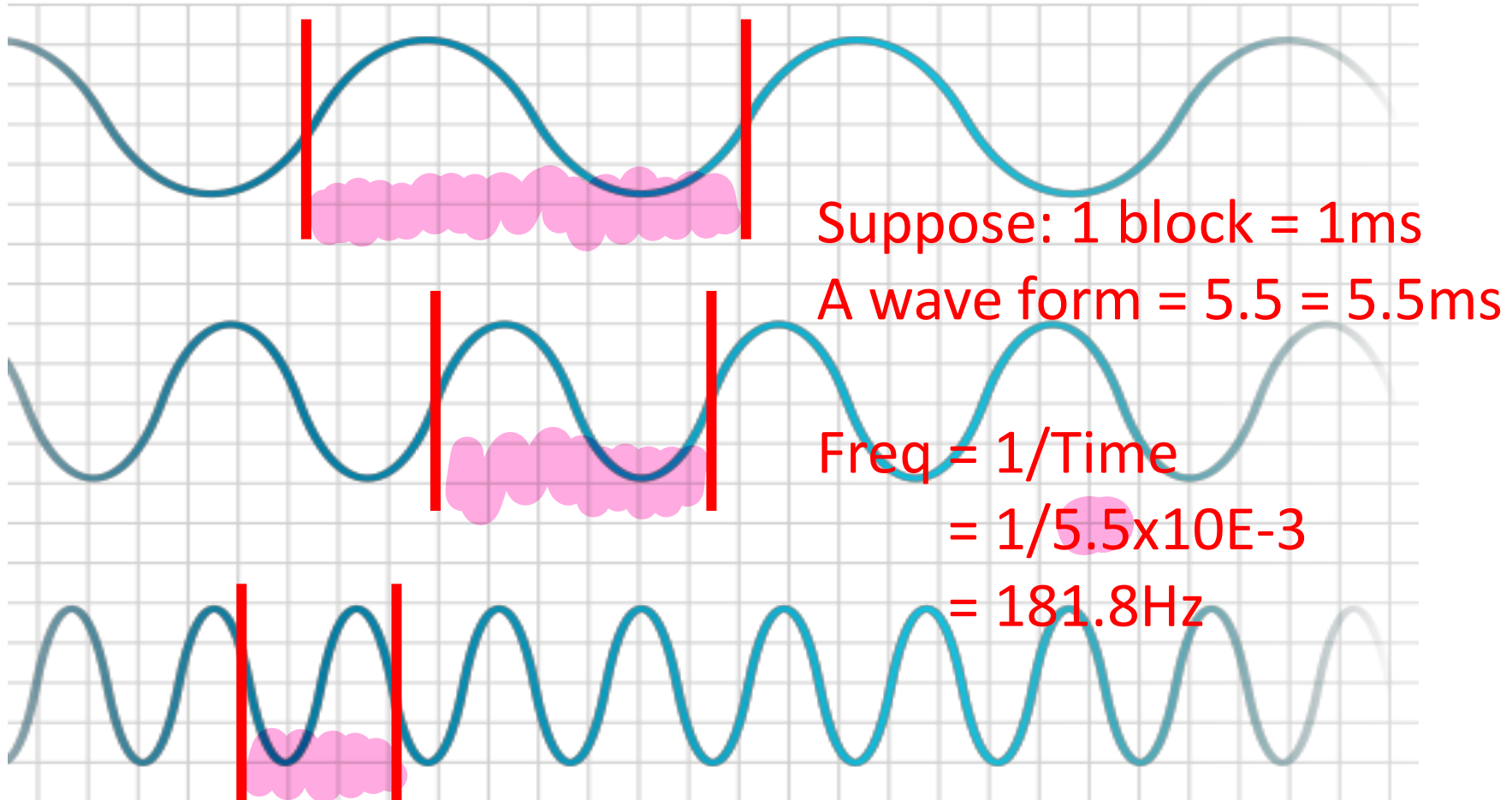
Which one is the low frequency?



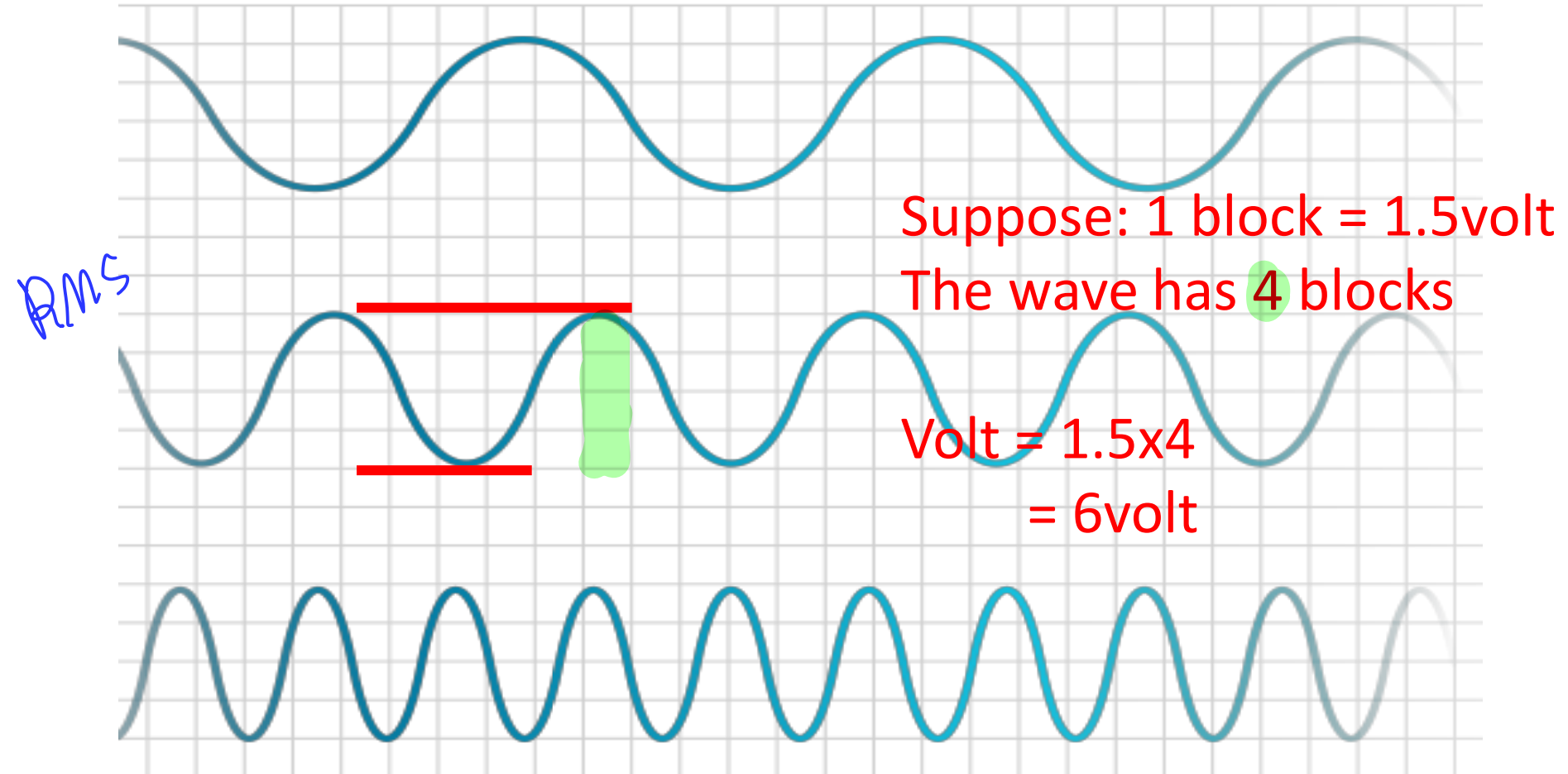
Which one is the low frequency?



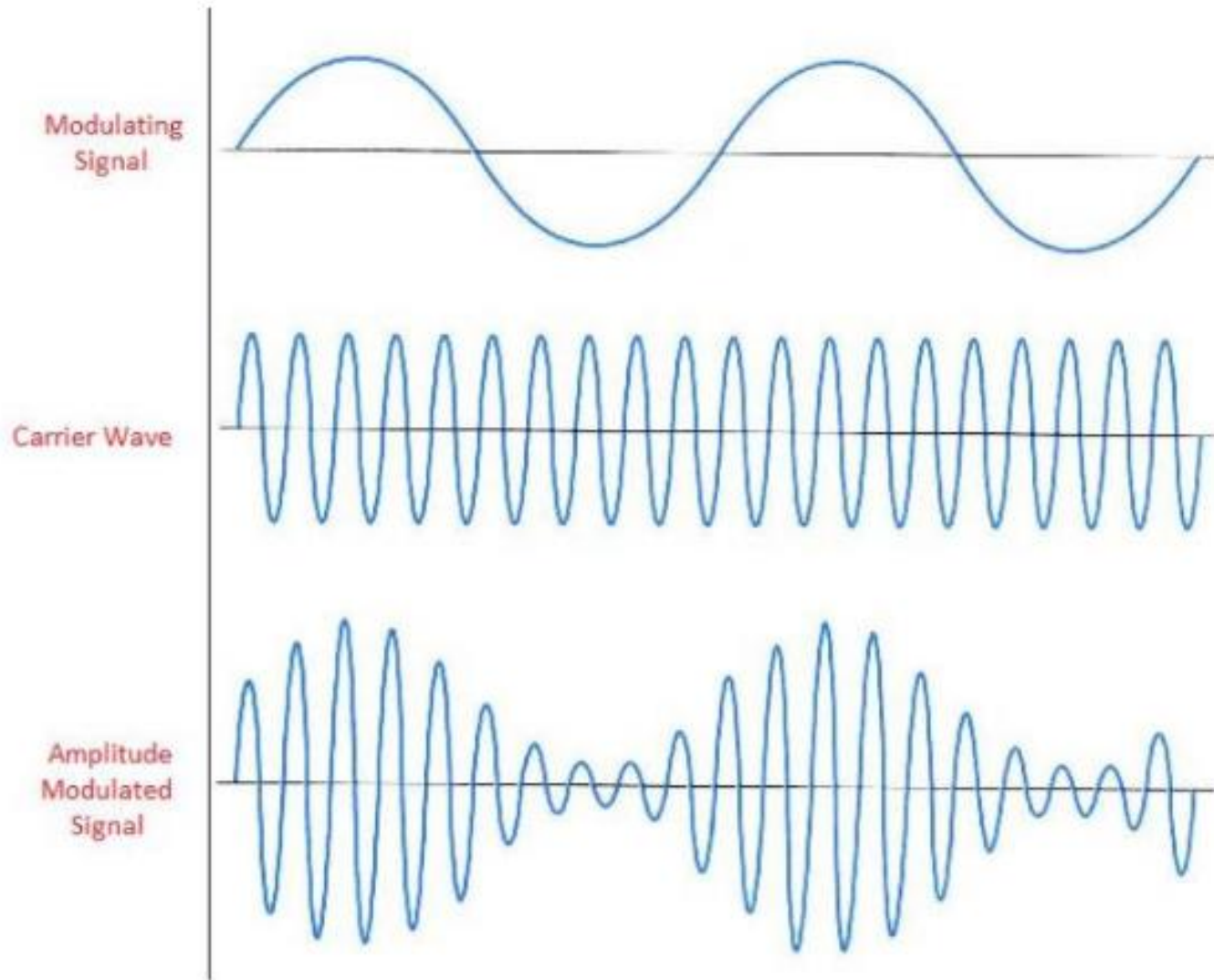
Activity 4.2 Frequency calculation



Amplitude, Voltage, Level



Modulation



A

+

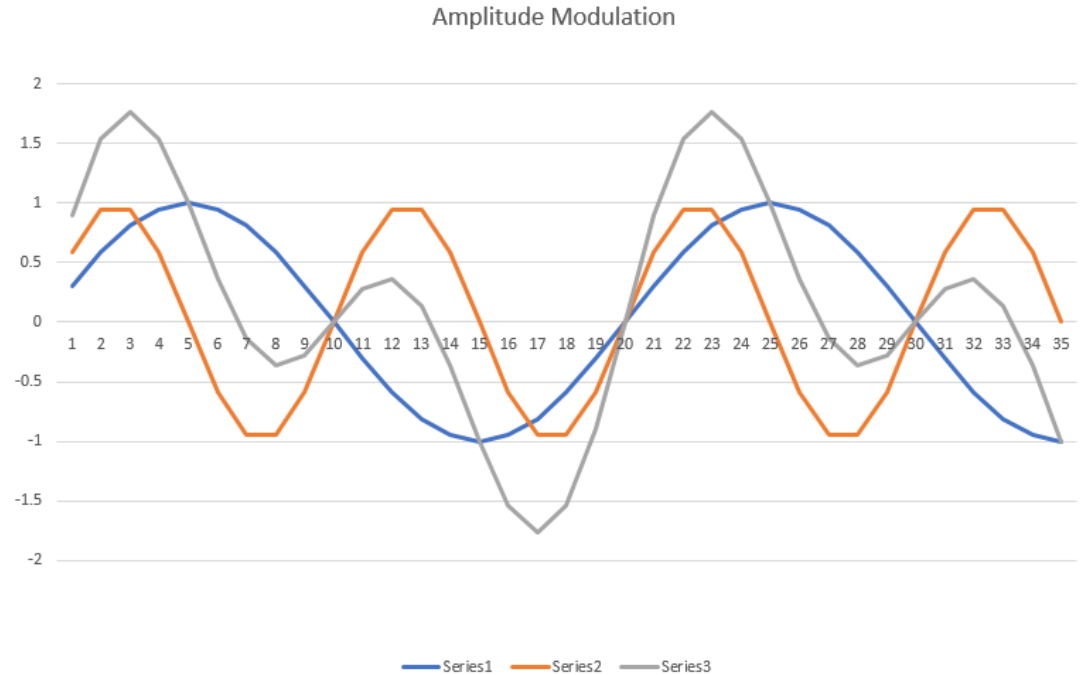
B

=

C

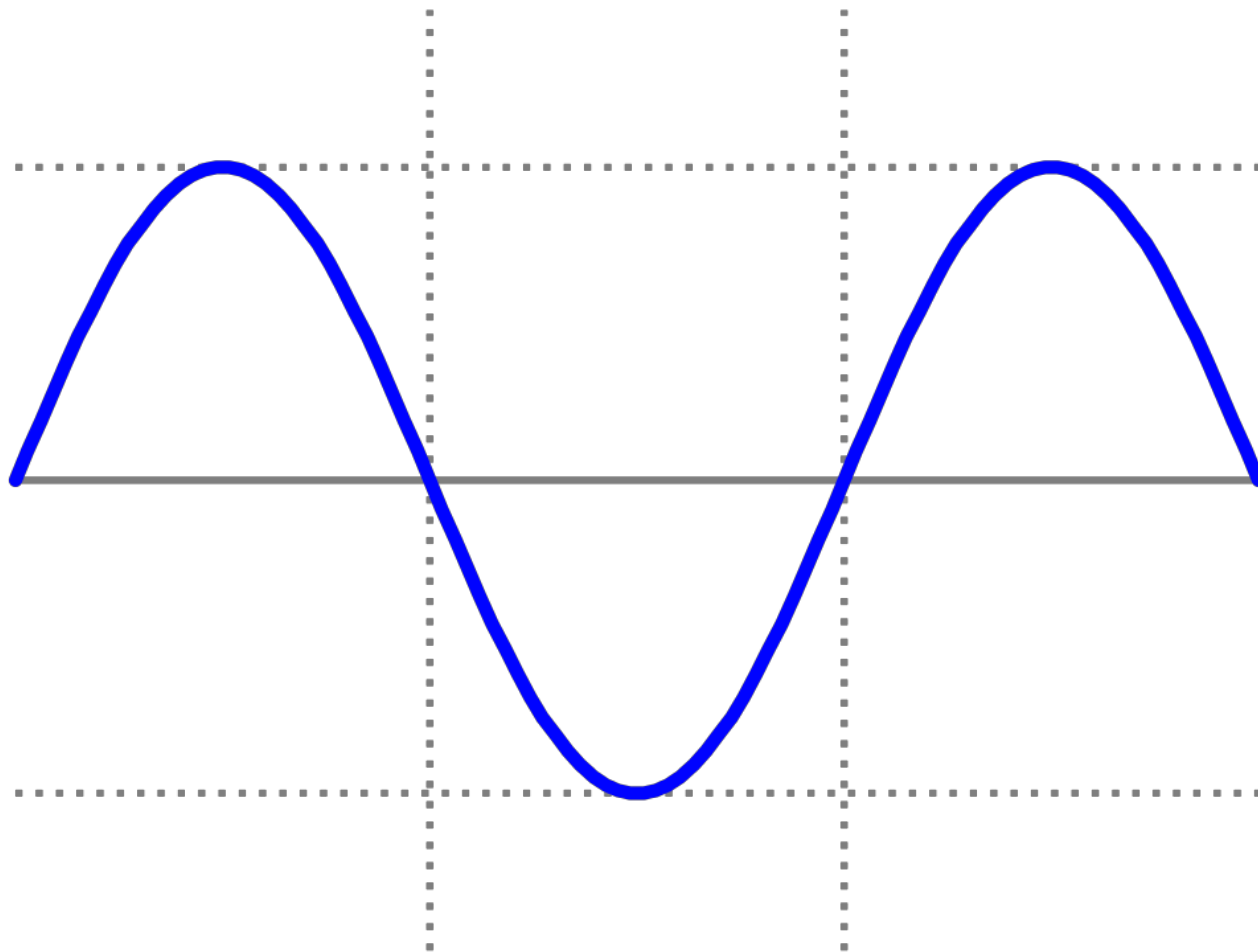
Activity 4.3 AM Simulation with Excel

	A		B		C
0.314159	0.309017		0.628319	0.587785	0.896802
0.628319	0.587785		1.256637	0.951057	1.538842
0.942478	0.809017		1.884956	0.951057	1.760074
1.256637	0.951057		2.513274	0.587785	1.538842
1.570796	1		3.141593	1.23E-16	1
1.884956	0.951057		3.769911	-0.58779	0.363271
2.199115	0.809017		4.39823	-0.95106	-0.14204
2.513274	0.587785		5.026548	-0.95106	-0.36327
2.827433	0.309017		5.654867	-0.58779	-0.27877
3.141593	1.23E-16		6.283185	-2.5E-16	-1.2E-16
3.455752	-0.30902		6.911504	0.587785	0.278768
3.769911	-0.58779		7.539822	0.951057	0.363271
4.08407	-0.80902		8.168141	0.951057	0.14204
4.39823	-0.95106		8.796459	0.587785	-0.36327
4.712389	-1		9.424778	3.68E-16	-1
5.026548	-0.95106		10.0531	-0.58779	-1.53884
5.340708	-0.80902		10.68142	-0.95106	-1.76007
5.654867	-0.58779		11.30973	-0.95106	-1.53884
5.969026	-0.30902		11.93805	-0.58779	-0.8968
6.283185	-2.5E-16		12.56637	-4.9E-16	-7.4E-16
6.597345	0.309017		13.19469	0.587785	0.896802
6.911504	0.587785		13.82301	0.951057	1.538842
7.225663	0.809017		14.45133	0.951057	1.760074
7.539822	0.951057		15.07964	0.587785	1.538842
7.853982	1		15.70796	6.13E-16	1
8.168141	0.951057		16.33628	-0.58779	0.363271
8.4823	0.809017		16.9646	-0.95106	-0.14204
8.796459	0.587785		17.59292	-0.95106	-0.36327
9.110619	0.309017		18.22124	-0.58779	-0.27877
9.424778	3.68E-16		18.84956	-7.4E-16	-3.7E-16
9.738937	-0.30902		19.47787	0.587785	0.278768
10.0531	-0.58779		20.10619	0.951057	0.363271
10.36726	-0.80902		20.73451	0.951057	0.14204
10.68142	-0.95106		21.36283	0.587785	-0.36327
10.99557	-1		21.99115	8.58E-16	-1



Convert Analog Signal to Digital Signal

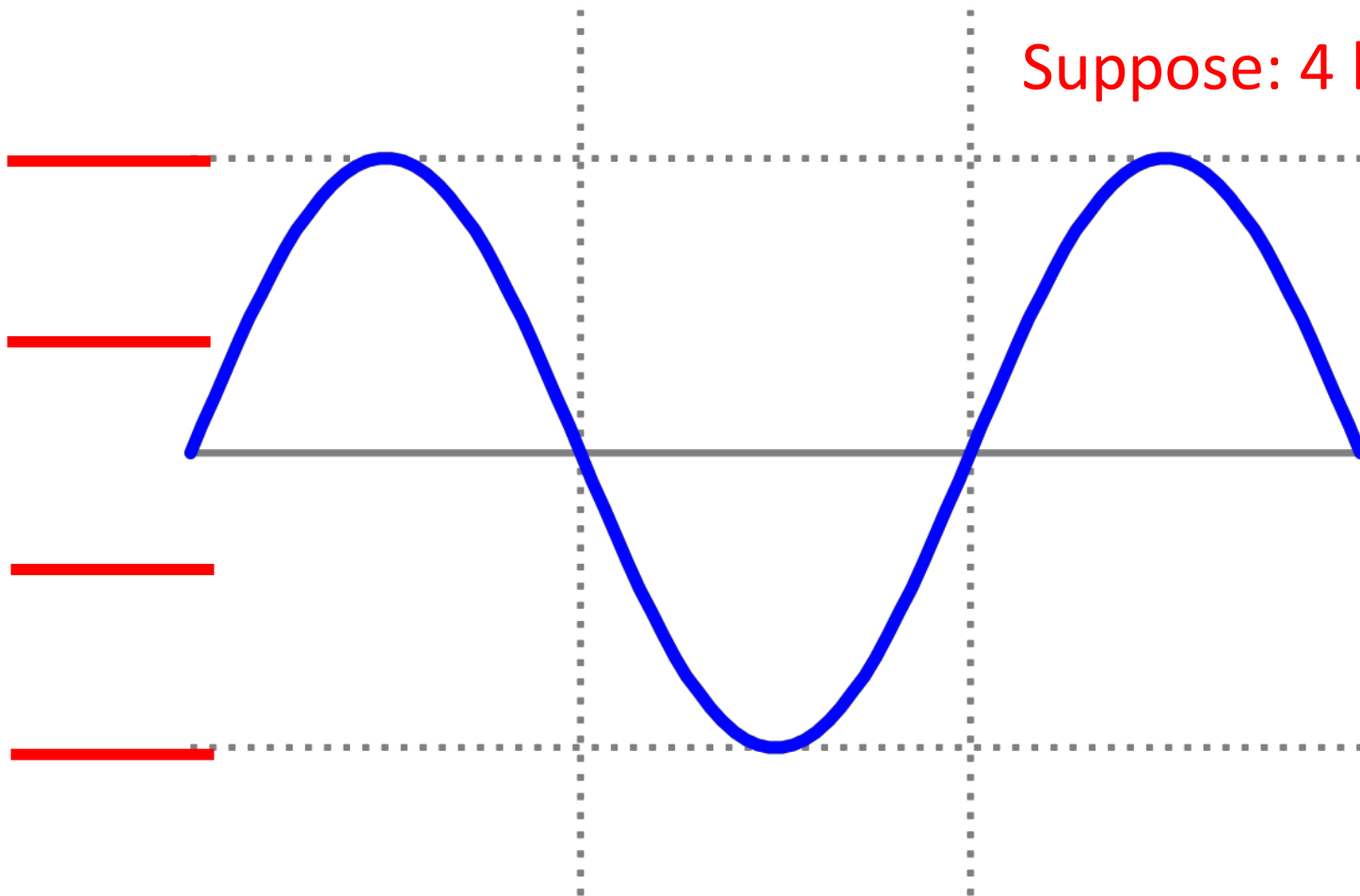
- Step1: Analog signal



Convert Analog Signal to Digital Signal

- Step2: Define the number of levels (bits)

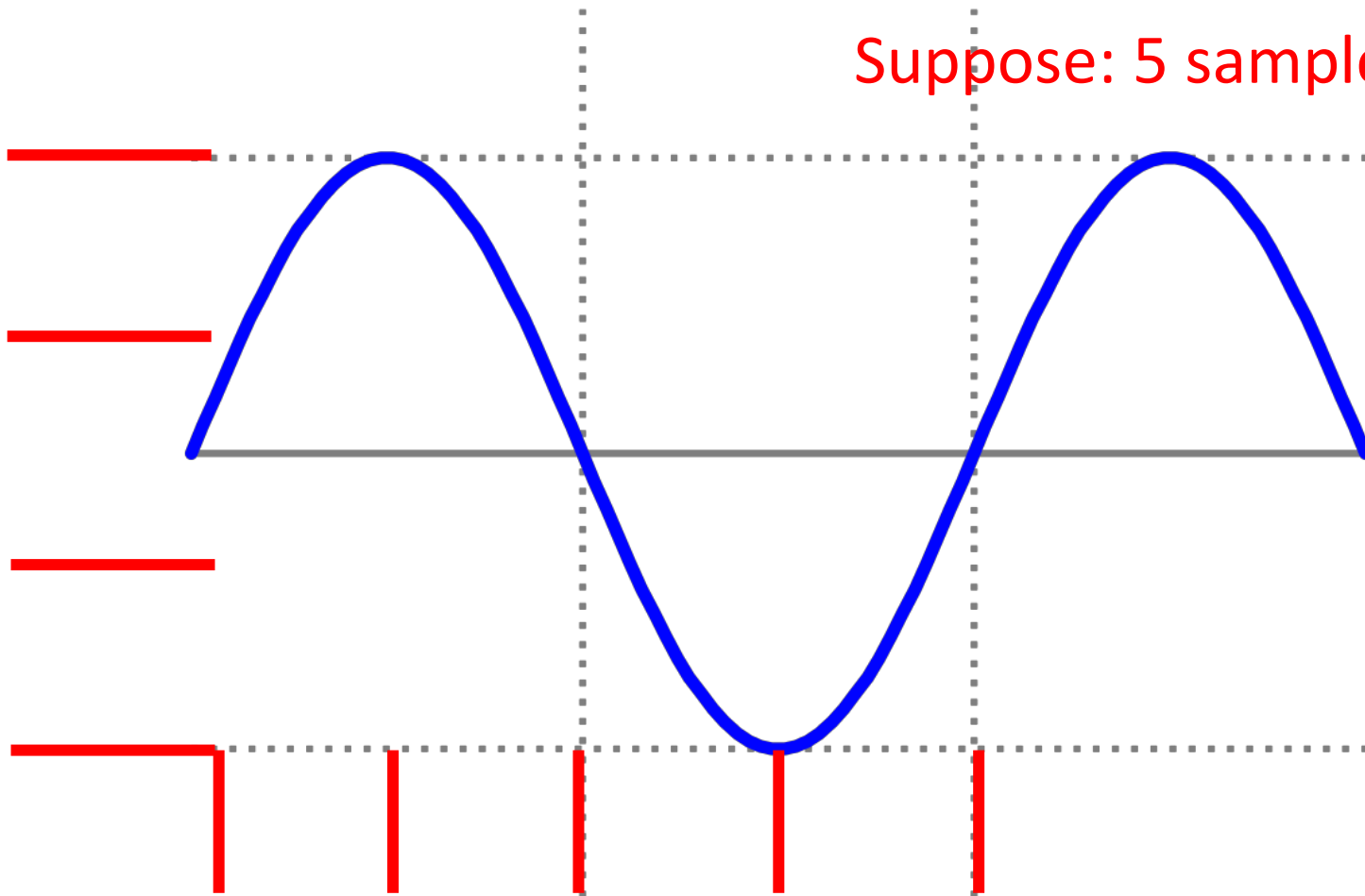
Suppose: 4 levels



Convert Analog Signal to Digital Signal

- Step3: Define the number of sampling

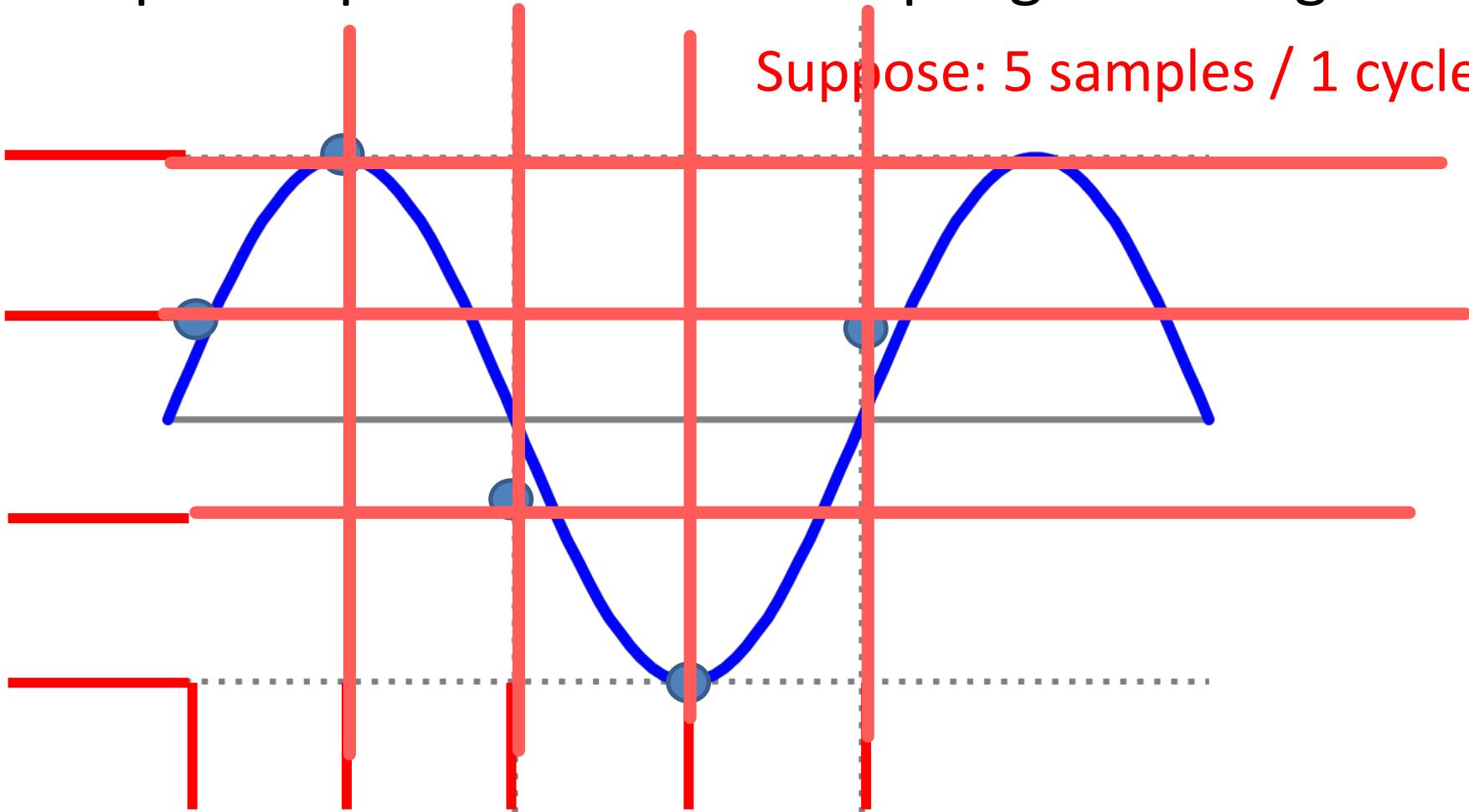
Suppose: 5 samples / 1 cycle



Convert Analog Signal to Digital Signal

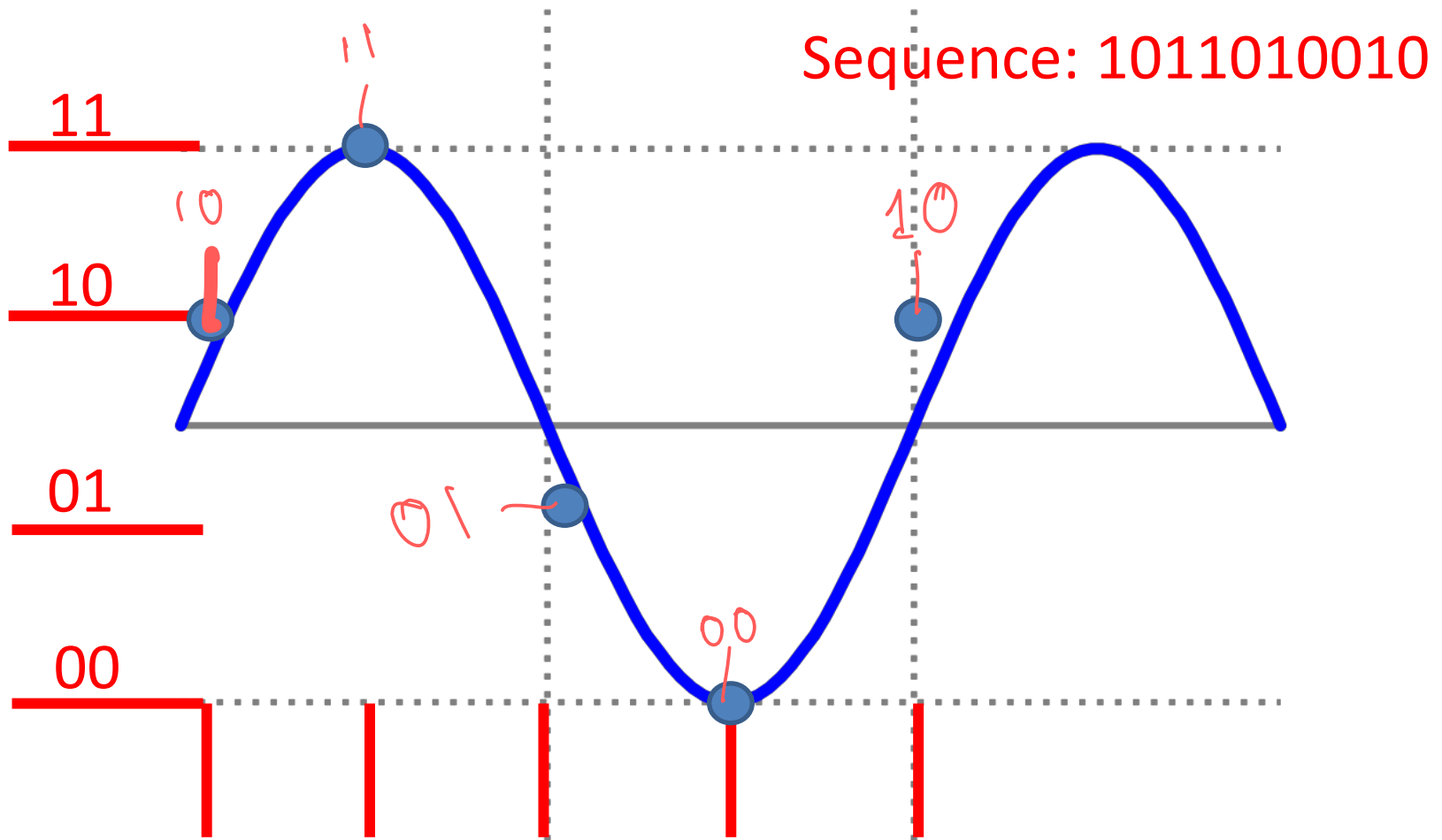
- Step4: Map the level and sampling to the signal

Suppose: 5 samples / 1 cycle

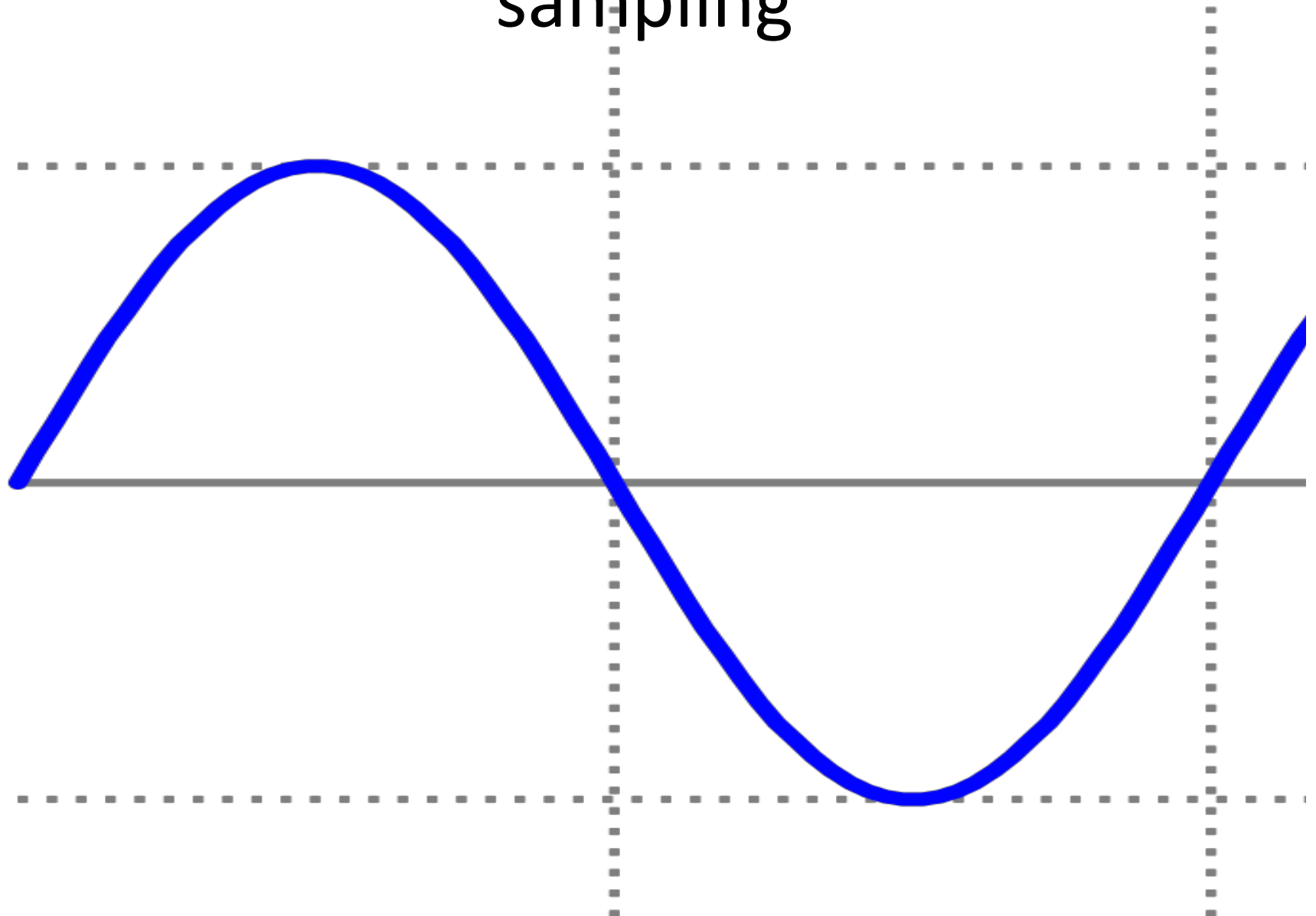


Convert Analog Signal to Digital Signal

- Step5: Reading digital signal



Activity 4.4 Drawing 8 levels and 10 sampling



Reference

- <https://www.quora.com/What-is-an-application-of-an-RS-flip-flop>
- <https://electronics.stackexchange.com/questions/jk-flip-flop-toggle>